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Applications of Lyapunov equation to derivative pricing

Applicazioni dell'equazione di Lyapunov al pricing di derivati finanziari

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Abstract (Italian)

Questo lavoro nasce da una collaborazione tra A2A, azienda leader in Italia nella fornitura di prodotti energetici, e l'università di Pavia. Scopo della collaborazione è lo sviluppo di nuovi strumenti per lo studio dei derivati finanziari.

Questo lavoro, in particolare, mira ad esplorare le possibili applicazioni della teoria dei sistemi lineari stocastici - nello specifico, dell'equazione di Lyapunov - al pricing di opzioni.

Il lavoro è suddiviso in quattro parti:

- La prima parte fornisce un'introduzione ai fondamentali della finanza. Vengono qui presentate le principali tipologie di derivati e viene introdotto il modello di Balck e Scholes per il pricing delle opzioni europee. Viene poi esposto il problema rappresentato dalla volatilità, l'unico dei parametri delle formule di Black e Scholes a non poter essere facilmente ricavato. Definita come la deviazione standard dei rendimenti di un titolo, la volatilità rappresenta una misura dell'incertezza sull'evoluzione del valore di un prodotto finanziario.

- La seconda parte esordisce con una rapida panoramica sui sistemi lineari stocastici. Questi rappresentano l'anello di collegamento tra la finanza quantitativa e la teoria dei sistemi dinamici. Nel caso il prodotto sottostante ad un'opzione possa essere modellato da un sistema lineare stocastico, la sua volatilità può essere calcolata mediante l'equazione di Lyapunov. Viene quindi selezionato un modello adatto agli strumenti scambiati sui mercati del settore energetico e viene applicata l'equazione di Lyapunov per ricavare una formula chiusa per la volatilità. Il risultato ottenuto viene poi inserito nelle formula di Black e Scholes e utilizzato nel pricing di opzioni. La capacità del modello così ottenuto di fornire prezzi in linea con il mercato viene provata mediante il confronto con un approccio più classico.

- La terza parte estende la teoria sviluppata nella seconda a modelli che contengono più di uno strumento finanziario. In particolare, viene dapprima affrontato il problema del pricing di una spread option, particolare opzione che deriva il suo valore dalla differenza tra i prezzi di due sottostanti. In seguito, viene sviluppato un approccio più generale, al fine applicare l'equazione di Lyapunov per ottenere la varianza di un portafoglio contenente un numero arbitrario di prodotti. Tale risultato viene poi impiegato nel calcolo del Valore a Rischio di un portafoglio.

- L'ultima parte è dedicata ad un riepilogo dei risultati ottenuti e all'analisi dei possibili sviluppi futuri.

I modelli e le procedure ricavate si sono dimostrate più efficaci di quelle precedentemente in uso in A2A, sia come capacità di fornire prezzi in linea con il mercato sia come efficienza computazionale. Di conseguenza, sono state integrate nei sistemi dell'azienda e sono attualmente utilizzate dal dipartimento Pricing and Risk Modelling.

Introduction

On 15 September 2008 Lehman Brothers, the fourth greatest American investment bank, declared bankruptcy, overwhelmed by a debt of $613 billion.

The biggest bankruptcy filing in the history of the United States left more than twenty thousand employees without a job and started the worst financial crisis the world has ever seen since World War II. From that day on, the words “market”, “option”, “derivative” have entered everyone’s dictionary.

Derivatives are usually considered mysterious and dangerous objects. The truth is way different. Not differently from every other tool, derivatives are made useful or harmful by their usage.

Surprisingly enough, non-financial institutions like energy company and airlines are among the most active traders of derivatives. Differently from financial firms, energy company are not interested in speculating on derivatives. These financial products act as insurances, protecting the business from unforeseeable events.

The need of A2A, one of the most important Italian energy companies, for more accurate instruments to study derivatives, gave birth to this work. The foundations of derivatives theory were laid in 1973, when the Black-Scholes model for pricing options was published. Options are among the simplest derivatives, yet among of the most common. After the work from Black and Scholes, an ever-increasing number of researches were carried out both by public institutions and private companies.

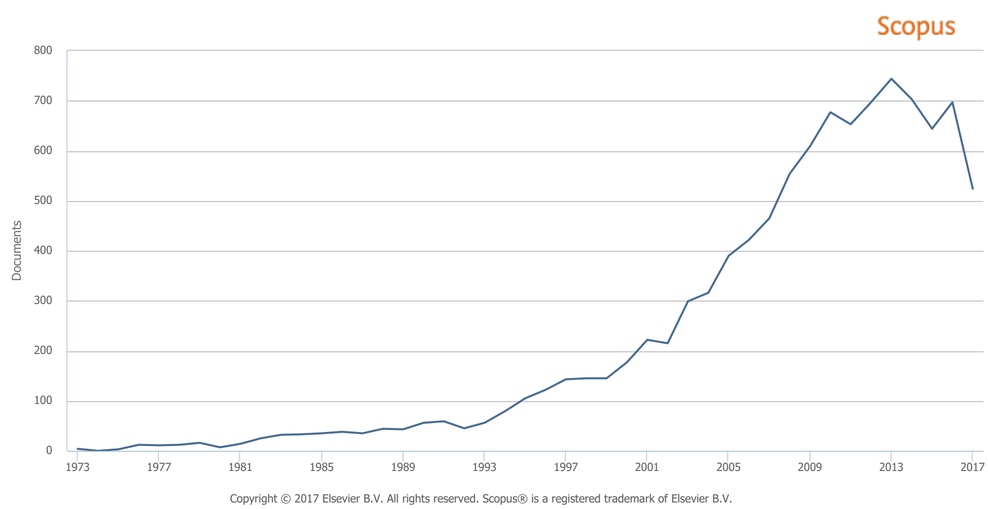


Figure 0.0.1: Number of papers matching the keywords ’Option pricing’ in Scoups, search engine by Elsevier.

By following the path of commonly accepted models and Nobel-prize-winning equations, this work wants to investigate some of the peculiar aspects of energy market with new instruments.

The markets of energy commodities - mainly electricity and gas - are relatively young. Italian electricity market, for instance, opened in 2004. The lack of a great amount of historical data and the low interest of big financial firms made most of the researcher focus on stocks markets. However, nowadays most industrialized countries have liberalized the energy sector and the need for specific studies is strong.

In collaboration with the Identification and Control of Dynamic Systems Laboratory of University of Pavia, A2A wants to explore the application of control system theory to the study of financial derivatives. Far from being merely academic, equations, formulas and algorithms born by the study are currently adopted by the company in its everyday operations.

This work is divided into 4 parts:

- Part I provides a basic overview of the most important financial concepts and introduces the Black-Scholes model, which is exploited in all the following parts. In the end, volatility is presented. Volatility is a measure of the uncertainty on the price of an asset and is the only parameter of the Black-Scholes equations which is very hard to estimate. This is where control theory comes into play.

- Part II introduces the fundamentals of stochastic linear systems. Stochastic systems are the bridge to link models for financial assets to control theory. If an asset can be modelled by a stochastic linear system, the Lyapunov equation can be solved to get a measure of its uncertainty. So, control theory may help in solving the problem of volatility. A model for energy commodities is selected, the Lyapunov equation is solved and the results plugged into the Black-Scholes formulas.

- Part III extends the ideas of Part II to models which involve more than one asset. The problem of pricing a spread option, which derives its value from two assets, is tackled. After that, focus is put on models involving an arbitrary number of assets. An analytical solution of the Lyapunov equation for multiple assets is derived. The result is then applied to the estimation of the Value at Risk of a portfolio.

- Part IV reviews the results of the work and suggest some future developments.

Part I

Background

Chapter 1

Markets, derivatives, options

Markets, derivatives, options: these words appear many times in the following pages. A good comprehension of their meaning is key to understand and, hopefully, appreciate, the whole work. The following paragraphs hence provide a very basic introduction to finance and its most elementary elements.

First of all, some historical background is presented: financial instruments are tools, each tool is best described by the needs it satisfies. Some modern and formal definitions of the most important concepts are then provided.

After that, focus is put on derivatives. Futures are first introduced, as they are the simplest form of derivatives. They are widely used, but they are not interesting to study as their price is fixed by economic laws.

Options then come into play. They are more difficult items, as they allow for a choice. A choice in the real word translates into a non-linear mathematical function in their model. Over the years a sophisticated theory for pricing options has been developed: however, before digging into it, it may be worth explaining why options are so common and how they are used.

Particular attention is devoted to the employment of options in the power market. Differently from financial firms, companies which act mainly in the energy sector are not interested in using options to gain money. They take advantage of derivatives to hedge the risk of adverse events which may impact both the collection of raw materials and their ability to deliver electric energy.

1.1 Historical Background: from the Middle Ages to Wall Street

1.1.1 The Middle Ages: bonds and market regulation

The practice of lending money and receive some interest was already known to the ancient Roman and Greeks. However, the first security which has ever been recorded was been issued on 12 March 1262 [14].

The city state of Venice was at war with Genoa and the Byzantine Empire and was struggling to raise the money it required to enhance its fleet. The Great Council then decided to issue a decree: whoever had borrowed some money to the state would have been awarded with a 5% interest until the complete refund of the debt.

This was, in fact, the first national bond.

Following the example of Venice, Genoa and Florence started to issue bonds to finance their army and their infrastructures.

Both private citizens and social institutions began to buy securities. And they started the business of trading them, too. In less than a century trading in government securities became so profitable that merchants started to apply what today we would call market manipulation. In 1351, Venice government had to outlaw spreading rumors intended to change the price of government funds.

The first market regulation had been born.

1.1.2 The Renaissance: shares and dividends

During the Renaissance the first market exchange was set up in Antwerp, Belgium. Initially it was nothing more than a building where merchants met to exchange bonds and securities.

When the epoch of major see voyages began, Dutch merchants realized that it was too risky for a single person to sponsor a trip. To decrease the risk of a shipwreck ruining their fortune, ship owners started to ask other merchants for money to support the trip and pay the crews. In case of a successful journey, the investor would have received part of the gain; in case of bad luck everyone would have lost his money. Shared risk, shared reward. These agreements, often lasting thing for one single trip, can be considered the first limited liability companies.

But the way business was done changed completely in the early XVII century, when the East India Companies was formed in Holland and England. To raise the money they needed to buy and prepare the ships, East India Companies started to sell part of themselves. Anyone could buy a certificate granting the ownership of a small piece of the company.

This is what today is called a share or a stock.

The possession of a share of the company gave the owner the possibility to get part of the profits of every single trip.

This is what today is called dividend.

Being written on paper the certificates could be bought and sold for money. At the very beginning trades took place in coffee shops, but it was just a matter of time for the first stock exchange to be created. Simply speaking, a stock exchange is a place where buyers and sellers can meet and trade stocks under commonly accepted rules. The London Stock Exchange opened in 1773, seventeen years before the New York Stock Exchange was formed. The latter is nowadays one of the most important stock exchanges in the world: its total capitalization overcomes the sum of London's, Tokyo's and Milan's.

Over the last two centuries, stock exchanges grew in importance and became more and more complex. Financial operations are now strictly controlled by authorities; bids are matched by algorithms rather then people; but the basic idea of stock and stock trading did not change.

1.1.3 From the ancient Greece to the New World: a short history of derivatives

The first trace of a derivative product can be found in Aristotle's Politics [27], where the Greek philosopher tells a story about Thales, the famous mathematician who lived in Miletus between 640 BC and 547 BC.

During the winter, he predicted that the olive harvesting during the following summer would be exceptionally large. Hence, he negotiated with the owners of olive presses the right, but not the obligation, to hire their facilities in the following summer. To grant himself this right, he made a cash deposit; however, due to the distance in time between the payment and the harvesting season, the price was considerably lower than the real value of press hiring. Eventually, it happened that the harvesting was indeed extraordinary and Thales was able to rent the presses to other farmers and make a fortune.

With the tale, Aristotle meant to show that any philosopher could easily become very rich, although this should not become an aim of theirs. What neither of the two philosophers did realize was that Thales had invented what we name today a call option. As the following paragraphs clarify, a call option is the right - not implying the obligation - of buying a specific asset, at a specific time, at a specific price.

It took two thousand years for the first derivative exchange to be set up. It happened in Osaka, Japan, during the XVII century . The Japanese city hosted the most important rice market in the country. As the market grew more and more, certificates of possession started to be traded in spite of the rice itself. The owner of the certificate received the goods with an agreed delay after the transaction took place. In the meantime, he could hold the certificate or sell it to another merchant.

That title represented what today is called a future.

In 1848, a first derivatives exchange was created in Chicago, United States. It was the Chicago Board of Trade (CBOT), the oldest organized derivatives market still operating in the world. However, the importance of derivatives exchange was relatively low with respect to stock markets until the end of World War II. In the following decades, the development of computer science and the consequent growth in available computing power led to the birth of more and more complex and articulated derivatives.

From the eighties onward, the diffusion of Over The Counter (OTC) contracts became considerable, OTC contracts being all deals which are agreed between two parties outside of a regular market.

In 2008, after the well-known Lehman Brothers bankruptcy, derivatives became notorious to the people. To clarify the meaning of some of the words which became buzzwords after the financial crisis, the following paragraph provides some formal definition.

1.2 An overview over financial instruments

To accomplish the non-trivial task of providing effective and authoritative definitions of the most important financial terms, two main sources are used: the Nasdaq online glossary [22] and Paul Wilmott's textbook On Quantitative Finance [32]. To ease the reading the sources are not referenced every time they are exploited.

Following a top-down approach, the first term to define is market. A financial market is an organized institutional structure or mechanism for creating and exchanging financial assets. Financial assets are divided into securities and real assets. Securities are pure financial instruments, contracts which grant some rights, while real assets, a.k.a commodities, are concrete, real-world objects like goods, buildings, pieces of equipment.

Securities can be further divided into three main categories:

Debt securities are usually released in exchange of a loan. They usually entitle the owner with the payment of interests until the loan is completely returned. Corporate and government bonds are the most representative examples.

Equities grant the ownership of a small piece of a company - and the right to receive a proportional part of its earning. They are also called shares or stocks.

Derivatives are all the securities whose value depends on the value of another asset. The simplest derivatives are futures and options, but more sophisticated instruments have been developed over the last decades.

The following diagram summarizes the different types of financial assets.

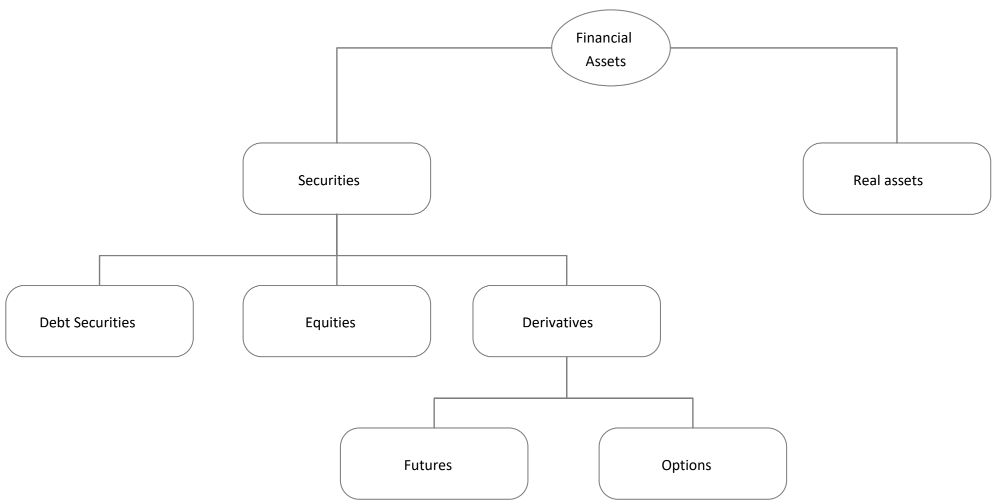


Figure 1.2.1: Types of financial assets.

The time has now come to dig a little bit deeper into the most widely spread types of derivatives: futures and options.

1.3 Futures and forwards

Before diving into the topic, a disclaimer is required.

Neither futures nor forwards are in the scope of this work: under some restrictive hypothesis, the classical economic theory provides a simple and widely accepted method to get their price; when those hypothesis are not fulfilled, the only force to drive the prices is the market itself.

However, a few lines about this simple kind of derivative make it easier to introduce the much more technical and sophisticated theory of option pricing.

Both futures and forward are contracts stating that one party will sell the other a predefined amount of a certain asset, on some fixed date in the future, at a given price. No money is exchanged until the contract expiration date. However, when that day comes, there is no choice for either of the parties. The terms of the agreement have to be fulfilled, irrespectively from the way the market behaved meanwhile.

The only difference between futures and forwards is the way they are exchanged. Futures are traded in regulated exchanges, while forwards are private agreements between two parties.

The previous explanation introduces to some terms which are widely used in derivative theory.

underlying is the asset to be exchanged when the contract expires. The value of the future naturally depends on its value.

maturity is the expiration date of the contract, when the underlying is delivered and money is paid.

spot price is the value of the underlying at a precise time. If no time is specified, then the price when the contract is traded is considered.

strike price is the money exchanged for the underlying at maturity.

Futures and forwards are particularly important both in the Oil & Gas and in the Power market: big industrial complexes, such as steel and chemical plants, require a huge amount of fuel and electric energy to operate. If the price of these commodities were subject to market fluctuations, the total costs for the company would be highly variable and quite unpredictable. This is why futures are bought by the facilities to cover their needs at fixed and certain costs. Futures on power supply can be traded both in dedicated exchanges and over the counter, but the latter mode is way more common.

1.4 Options

While a future carries obligation, an option let the owner make a choice. Two basic kind of options exist:

call options give the right to buy a specific underlying for a fixed strike price, at a given time in the future (maturity)

put options entitle the right to sell a specific underlying at a fixed strike price, at a given time in the future

As they are the most common types, they are called vanilla options, in analogy with the ice-cream flavour, widely spread in the United States. All the other types of options are called exotic. Both exotic and vanilla options may exist in different variants, depending on when the rights on the underlying can be exercised. The main groups are:

American options let the owner exercise its right at every time before maturity

European options make the underlying be exchanged only at maturity

As American options carry further complications, due to the possibility of early exercise, from now on only European options are considered.

1.4.1 A mathematical model for options: payoff

When a European call option comes to maturity, the owner has to choose whether to exercise the right of buying the underlying. However, the choice is driven by the spot price of the underlying at maturity. If it is higher than the strike price, it is convenient to exercise the option: this way it is possible to buy the underlying at strike price and sell it to the market at an higher price, thus achieving a net profit. Otherwise, it is wiser to let the option expire without taking any further action.

Following these considerations, it is possible to define the payoff of a call option as a function of the strike K, the time-varying spot price $S(t)$ and the maturity T:

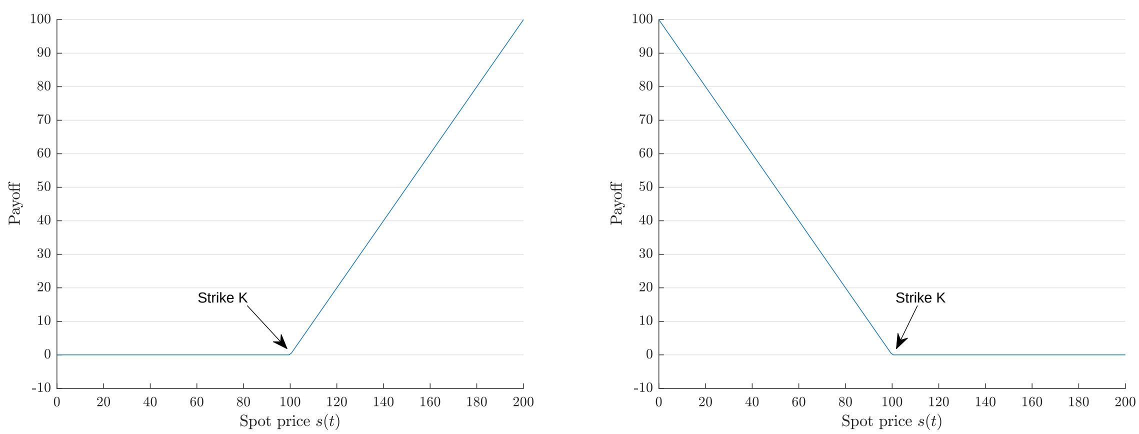
$$P_c(K, s, T) = \max(S(T) - K, 0) \tag{1.4.1}$$

The non-linear function max accounts for the choice the option allows.

If a put option is considered, the exercise is profitable if the spot price at maturity is lower than the strike. If this is not the case, it is more convenient to sell the underlying to the market. The payoff for a put option is thus given by:

$$P_p(K, S, T) = \max(K - S(T), 0) \tag{1.4.2}$$

Payoffs of European vanilla options are depicted in the following chart.



(a) Call option. (b) Put option.

Figure 1.4.1: Payoff functions for vanilla option.

1.4.2 Spread options

Spread options derive their value from the difference in price between two underlyings.

In the most general case, a call spread option gives the right to buy a unit of a first underlying with a unit of a second underlying and a certain amount of cash. The payoff of such an option is thus given by:

$$P_{sc} = \max(S_1(T) - S_2(T) - K, 0) \tag{1.4.3}$$

A put spread option, on the other hand, entitles the owner with the right to sell a unit of a first underlying and receive as a compensation a unit of a second underlying and a certain amount of cash. The payoff of a put spread option is:

$$P_{sp} = \max\left(S_2\left(T\right) + K - S_1\left(T\right), 0\right) \tag{1.4.4}$$

When K equals zero, spread options actually give the right to exchange the two underlyings.

If an investor buys a spread option with $K = 0$, and payoff $\max\left(S_1\left(T\right) - S_2\left(T\right), 0\right)$, he wants the price of asset 1 to increase over the price of asset 2, in order to exercise the option and gain the difference between the values of the two underlyings.

1.5 Why trading options: speculation and hedging

There are plenty of reasons why an investor may want to trade options. However, most of them fall into two main groups: speculation and hedging.

Speculation is about taking a risk. The ones who speculates usually pretend to have a clear vision of how the market will behave within a certain time horizon. They invest according to their vision and they can either loose or gain a lot of money, depending on whether their forecast is correct or not.

To understand how options are used by speculators, an example is now provided.

Let $S\left(t\right) = \$100$ be the price of some stock at the current date t and $O = \$10$ the market price for a call option with strike $K = \$120$ and maturity T. The investor believes that the price of the stock will raise well over K by maturity. He wants to risk his capital on his vision: he only has to choose whether to buy call options or the stock itself.

Suppose he opted for the stocks and pretend he was right: the spot price of the underlying at maturity is $S\left(T\right) = \$150$. His total gain, in percentage, is:

$$\frac{S\left(T\right) - S\left(t\right)}{S\left(t\right)} = \frac{\$150 - \$100}{\$100} = 0.5 = 50\% \tag{1.5.1}$$

Now keep the previous scenario but suppose he invested in call options. The return becomes:

$$\frac{S\left(T\right) - K}{O} = \frac{\$30}{\$10} = 3 = 300\% \tag{1.5.2}$$

which is way more than the profit he could realize from the stocks.

However, suppose now the investor was unlucky: the price at maturity turns out to be $S\left(T\right) = \$110$. If he bought stocks, he would still be profitable, as:

$$\frac{S\left(T\right) - S\left(t\right)}{S\left(t\right)} = \frac{\$110 - \$100}{\$100} = 0.1 = 10\% \tag{1.5.3}$$

If he opted for options, instead, he would loose all his money, as a call option is worth nothing as long as the price of the underlying is less than the strike.

Using options for speculation is extremely risky: rewards can be high, but losses can be huge as well.

On the other hand, hedging is about mitigating risk.

Hedging means entering some financial contract to offset the risk of another investment. The text-book example of hedging through option usage is provided by airlines.

Airlines know the have to buy fuel as long as they want to stay in the business. However, fuel price is influenced by oil price and refining capacity. These factors are driven by politics, natural events and other exogenous factors which are mostly unforeseeable. To mitigate the risk of a high fuel price impacting too much on the costs of the flights, air companies buy a great amount of call options on jet fuel from oil companies. If the fuel price skyrockets, air companies use the options to buy raw materials at a fixed price. This way, the derivatives act like an insurance for airlines.

However, no insurance is free. An option itself has a cost. Moreover, if the fuel price decreases, it is more convenient for the company not to exercise the option and buy the commodity from the market. If this is the case, the price paid for the option is lost without bringing any advantage.

1.6 Options in power markets

The more a market is uncertain and variable, the more hedging is required. Electricity price is known to be very volatile, meaning it can face fast and unforeseeable variations in a very short time range. The main indicator of the variability of an asset price is volatility, whose formal definition is given in Chapter 2. Previous studies [30] found the following results:

- Volatility of a government bond is usually lower than 0.5%
- Volatility of a stock lies between 1% and 4% in most cases
- Volatility of electricity prices can overcome 50%

This is mainly due to the fact that there is currently no way to effectively stock a large amount of electric energy: the offer / demand balance has thus a great impact on prices. In order to hedge the risk related to the price of electricity, large industrial facilities are interested in buying both futures and options - just like airlines in the previous section. However, energy company themselves trade derivatives.

To better understand why this happens, two scenario are considered. Suppose that, for any reason, the price of electricity falls down, under the break-even point of a certain company. If that company had put options with higher prices, it could exercise them and mitigate the losses.

On the other hand, imagine a company suffering from a failure in one of its power plants. Its productive capacity is heavily impacted: nevertheless, contractual obligation with the customers have to be fulfilled. Moreover, the loss of production may lead to an increase in market prices. However, if the unlucky company owns some call options issued by another firm of the same sector, electricity can be bought at a fixed price, regardless what the market price becomes.

Besides futures and vanilla options, more sophisticated derivatives are used by energy company. Spread options, for example, are very common.

Most of the biggest energy companies operate in different countries. To carry gas or electric energy, they buy transportation capacities from the owner of the pipes or from the power grid manager. Once they have bought the capacity, they have to choose whether to actually move their product from one country to another or not. This is by all means comparable to the ownership of a spread option, the underlyings being the prices of the good in the two countries and the constant K the transportation costs.

When oil and gas markets are considered, the situation is not very different. Despite being less volatile than electricity because of the mitigating action of stocks, oil and gas are mostly produced by a reduced number of countries. The price of these commodities is thus exposed to political and natural events.

Only few of many possible usages of options by energy companies have been presented: lots of papers provide a far more exhaustive overview - see, for example, [26].

Following the previous considerations, it is not difficult to realize that energy companies trade a huge amount of derivatives. In 2016, Eni reported €482 million losses due to derivative trading, while Edison declared about €10 billion of total derivatives either held or sold by the company.

In such a scenario, the problem of finding a "fair" price for an option becomes crucial.

Chapter 2

Pricing options: the Black-Scholes formula

Options can be profitable or not, surely they have no downside. The minimum payoff the owner can achieve is zero, meaning that he can never lose money. This is why an option is valuable by itself. The question is: what is the value of an option?

This is where mathematics modelling comes into play. The classical, widely-accepted formula for options pricing is based on a stochastic model of the underlying.

As all the models, it holds under some assumptions and, even when these assumptions are verified, it is nothing more than an approximation of the real world. In the following pages, the main steps of the derivation of the Black-Scholes formula are presented.

To better introduce the very idea of pricing, futures are first considered. Pricing future is much simpler than pricing options: for this reason future pricing theory is very useful to introduce the fundamental ideas derivative pricing is based on.

After some further clarification on the concept of pricing and its difference with forecasting, a stochastic model for the underlying is presented. Assuming it well describes the actual behaviour of the underlying, the same principles adopted to price futures are applied. The result is a closed-form formula of the price of a European vanilla option.

The Black-Scholes formula, published in 1973 [3], was a major breakthrough is quantitative finance. Twenty-four years later, two of the authors, Myron Scholes and Robert Merton, were awarded of the Nobel prize for Economics - Fischer Black died two years before.

In the end, the problem of volatility is introduced.

Volatility is the one and only parameter of the Black-Scholes formula which is not established by the option contract and cannot be easily inferred from the market. Classical methods of estimating volatility are reviewed, thus paving the way to the main topic of this work.

2.1 An easy start: pricing futures

Interest rates and the no-arbitrage principle are the fundamental blocks the classical pricing theory is built on.

Under the assumption that the underlying can be supplied without restrictions, the “right price” for a future can be derived from the no-arbitrage principle: no profit can be made without taking risks higher than the interest rate on cash.

The interest rate, also known as the risk-free interest rate, models the gain one can achieve by simply putting some cash in a bank. This gain is assumed to be guaranteed and riskless.

Let r denote the interest rate and x_i the amount of money in the bank at time i. After one time unit, the amount of money becomes:

$$x_{i+1} = x_i (1 + r) \tag{2.1.1}$$

and after t time units:

$$x_{i+t} = x_i (1 + r)^t \tag{2.1.2}$$

Suppose now the interest to be paid in m instalments of amount r/m, one for each time unit.

$$x_{i+t} = x_i \left(1 + \frac{r}{m}\right)^{mt} \tag{2.1.3}$$

If m tends to infinity, then:

$$x_{i+t} = \lim_{m \to +\infty} x_i \left(1 + \frac{r}{m}\right)^{mt} = x_i e^{rt} \tag{2.1.4}$$

So, e^{rt} is the maximum profit which can be made without taking risks.

When a future contract is entered, the following quantities are known:

- The time when the contract is signed t
- The maturity T
- The spot price $S(t)$
- The strike F

To ease the following computations, interest rate r is supposed constant and transaction costs are neglected.

What is still unknown is the spot price at maturity $S(T)$. However, by creating a special portfolio, it is possible to nullify the impact of the uncertainty on the price at maturity.

At time t, a future contract to buy a certain amount of the underlying is signed. At the same time, the same amount of the asset is sold. Selling an asset without owning it is called short-selling: it is allowed by most markets under some time constraints on the delivery of the asset. This transaction leads a net gain of $S(t)$ in cash, which is put in the bank. At maturity, the strike price F of the future is paid and the underlying is used to close the short-selling. So, the strategy guarantees a loss of F from the future and a gain of $S(t)\, e^{r(T-t)}$ from the interests on the bank account. Since the initial investment in the portfolio was zero, the value at time T has to be the same. Thus:

$$-F + S(t)\, e^{r(T-t)} = 0 \tag{2.1.5}$$

where F takes a negative sign because it is a loss. From the previous equation, the value of the forward contract can be derived:

$$F = S(t)\, e^{r(T-t)} \tag{2.1.6}$$

If the price were different from this value, riskless profit opportunities would arise. For example, in case $F < S(t)\, e^{r(T-t)}$, one could build the same portfolio previously described and get a net profit at maturity:

$$-F + S(t)\, e^{r(T-t)} > 0 \tag{2.1.7}$$

The standard economic theory states that, if such a situation happened, investors would act to exploit the gain opportunity. They would buy the futures, increasing the demand and thus the price, until equilibrium is met again.

2.2 An important distinction: pricing vs forecasting

The pricing model for futures is oversimplified and limited: among other factors, it does not take into account transaction costs and variable interest rates.

However, it is a good example of what pricing is about. The value of F derived from the no-arbitrage principle does not necessarily match the market price of the future - which can be influenced by exogenous factors. Nor it guarantees zero profit at maturity, if only the future is bought - the profit depends on $S(T)$, which is unknown.

It is only a "fair price" which guarantees that no strategy can be elaborated at time t to make certain net profit.

Pricing is thus intrinsically different from forecasting. While the latter tries to foresee the price of an asset in the future, the former wants to establish a price such that, with the information available today, no arbitrage is allowed.

2.3 A stochastic model for assets

2.3.1 A discrete-time model

Consider a stock whose price is $S(t_i)$ at some instant $t = i$. Assume the difference in price of the stock follows a random behaviour: it is normally distributed with zero mean. That is:

$$S(i+1) - S(i) \sim N\left(0, \sigma^2\right) \tag{2.3.1}$$

where $N\left(\mu, \sigma^2\right)$ denotes a Gaussian distribution with μ mean and σ^2 variance. An important question is: how does the variance scale with time?

The difference in price of the stock between t_2 and t_0 is:

$$S(i+2) - S(i) = [S(i+2) - S(i+1)] - [S(i+1) - S(i)] \sim N\left(0, \sigma^2\right) + N\left(0, \sigma^2\right) \tag{2.3.2}$$

Assume now the variations in different time ranges are independent. Then:

$$S(i+2) - S(i) \sim N\left(0, 2\sigma^2\right) \tag{2.3.3}$$

The same argument can be applied to any time range: the variance of the returns scales linearly with time. If one defines $\Delta S = S(i + \Delta t) - S(i)$ and z as a standard Gaussian random variable $N(0,1)$, the time variation of a stock price follows the law:

$$\Delta S = z \cdot \sigma\sqrt{\Delta t} \sim N\left(0, \sigma^2 \Delta t\right) \tag{2.3.4}$$

A discrete random process which follows the previous rule and has $\sigma^2 = 1$ is called a discrete-time Wiener process.

2.3.2 A continuous-time model

By taking the limit for Δt approaching zero and keeping the same properties stated in the previous section, one can define the continuous-time Wiener process.

Definition 1 (Wiener process).
A Wiener process is a stochastic process W with the following properties.

1. $W(0) = 0$ *almost surely - i.e. with probability 1.*
2. *Increments of W are independent from the past history: $W(t + \Delta t) - W(t)$ is independent from $W(s)$ for each $t > 0$, $\Delta t > 0$ and $s \leq t$.*
3. *Increments of W are normally distributed, with zero mean and variance which scales linearly with time: $W(t + \Delta t) - W(t) \sim N(0, \Delta t)$ for each $t > 0$ and $\Delta t > 0$.*
4. *W has almost surely continuous paths.*

It is possible to generalize the previous definition by adding a deterministic trend. A generalized Wiener process (a.k.a arithmetic Brownian motion) X follows the law:

$$dX = \mu dt + \sigma dW \tag{2.3.5}$$

The infinitesimal increment dX is thus the sum of the deterministic increment μdt and a stochastic noise σdW, dW being an infinitesimal increment of a Wiener process. A realization of a Wiener process is compared with a realization of a generalized Wiener process in the following chart.

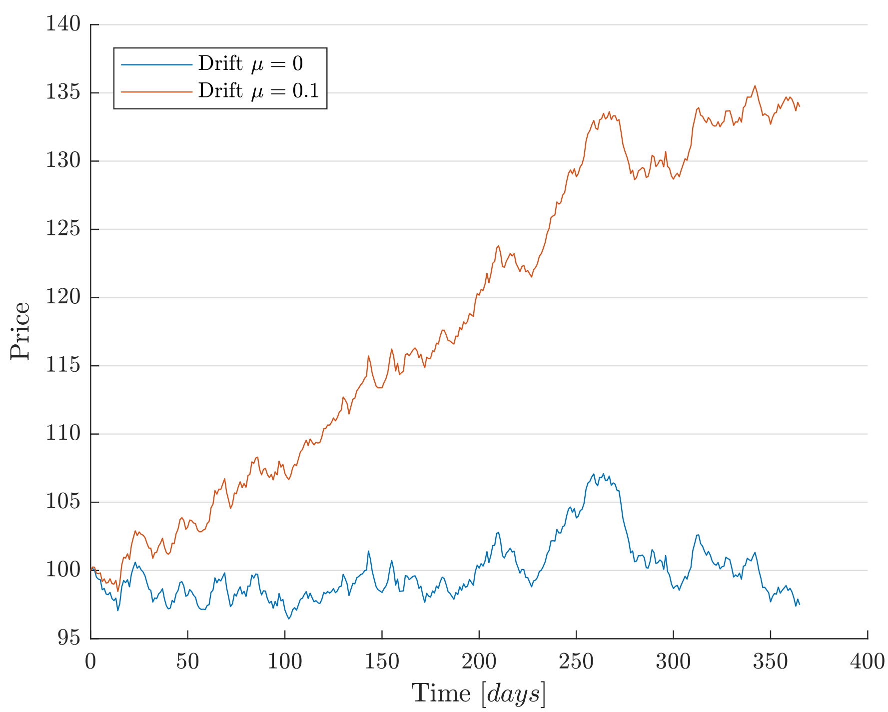


Figure 2.3.1: Wiener process and generalized Wiener process.

However, a generalized Wiener process is not a good model for the price of a stock for two reasons.

1. Its value $X(t)$ may become negative, while a price cannot
2. Both the deterministic trend and the stochastic variation are not influenced by the current price of the stock. This is unrealistic, as investors care about the relative return of their investment. If the reward required to buy a \$10 stock is 5%, then the reward required to buy a \$50 stock would be 5% as well, all the other conditions being the same.

To fix the two problems, it is sufficient to model the relative variations rather than the absolute ones. This leads to the definition of another stochastic process, named geometric Brownian motion:

$$\frac{dS}{S} = \mu dt + \sigma dW \tag{2.3.6}$$

which has a differential which is proportional to the value of the process itself.

$$dS = \mu S dt + \sigma S dW \tag{2.3.7}$$

The different behaviour of geometric Brownian motion and Wiener process in clearly highlighted by the following chart.

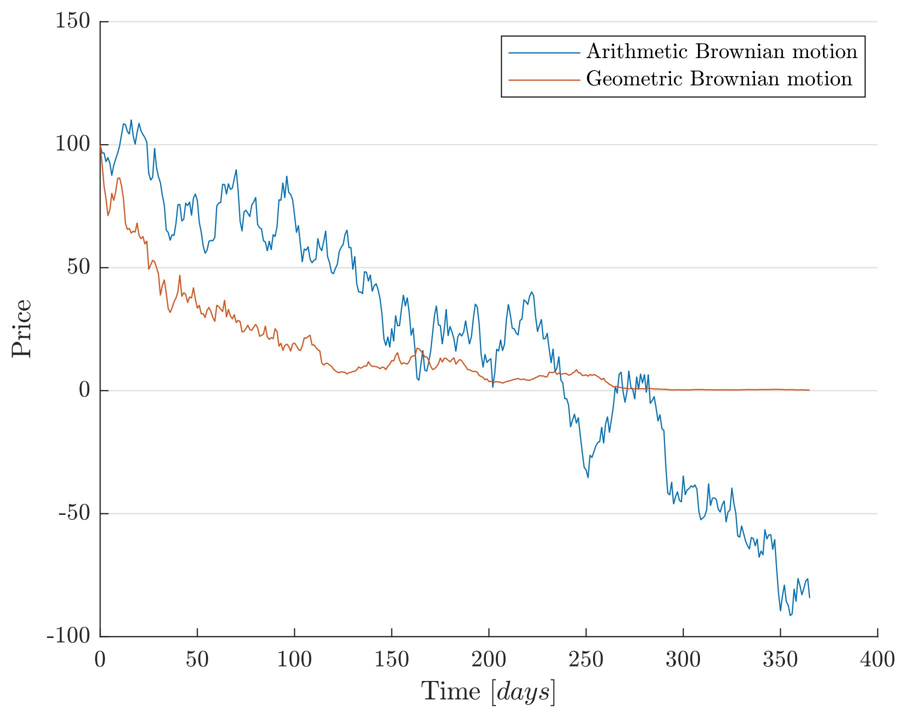


Figure 2.3.2: Generalized Wiener process and geometric Brownian motion.

Even the Geometric Brownian motion fails to model two aspects.

- Geometric Brownian motion is by definition continuous, while stock prices may not
- Geometric Brownian motion has a constant volatility, while stocks return standard deviation changes in time

Despite these downsides, Geometric Brownian motion is still considered a good model for stock prices.

A more exhaustive dissertation on the topic, together with a justification of the assumptions introduced at the beginning of this section, can be found in [32] and [17].

2.3.3 Log-spot prices and returns

One last step has to be made. It may be useful to know how the natural logarithm of the stock price behaves. To do so, the following theorem should be introduced.

Theorem 1 (Ito's Lemma).
Let W be a Wiener process, a and b deterministic functions and Y an Ito's process - i.e. a stochastic process driven by:

$$dY = a\left(Y,t\right)dt + b\left(Y,t\right)dW \tag{2.3.8}$$

Let $G\left(Y,t\right)$ be a function of the Ito's process Y and the time t. Then

$$dG = \left(\frac{\partial G}{\partial Y}a + \frac{\partial G}{\partial t} + \frac{1}{2}\frac{\partial^2 G}{\partial Y^2}b^2\right)dt + \frac{\partial G}{\partial Y}b \cdot dW \tag{2.3.9}$$

One should then define:

$$X = \ln S \tag{2.3.10}$$

and apply Ito's lemma to Equation 2.3.7. The result is:

$$dX = \left(\mu - \frac{1}{2}\sigma^2\right)dt + \sigma dW \tag{2.3.11}$$

which is again a generalized Wiener process, a.k.a Arithmetic Brownian motion.

This fact implies that the log-spot prices $X\left(t\right)$ are normally distributed and their variance, at any time t, is $\sigma^2 t$. Moreover, spot prices $S\left(t\right) = e^{X(t)}$ have a log-normal distribution. This is coherent with the requirements, as a log-normal variable cannot assume negative values.

It may be worth noting that the log-spot prices X are closely connected to the returns.

Returns R are formally defined as the relative variation of an asset price:

$$R = \frac{S\left(t+\Delta t\right) - S\left(t\right)}{S\left(t\right)} \tag{2.3.12}$$

As Δt tends to zero:

$$R \simeq \ln\left(1+R\right) = \ln\left(1 + \frac{S\left(t+\Delta t\right) - S\left(t\right)}{S\left(t\right)}\right) = \ln\left(\frac{S\left(t+\Delta t\right)}{S\left(t\right)}\right) = \ln\left(S\left(t+\Delta t\right)\right) - \ln\left(S\left(t\right)\right) \tag{2.3.13}$$

Modelling the log-spot prices is thus equivalent to modelling the returns.

2.4 The Black-Scholes model

2.4.1 Assumptions: the Black-Scholes world

Assumptions of the so-called Black-Scholes world are now introduced.

- The underlying price is ruled by a geometric Brownian motion.
- The risk-free interest rate is constant.
- The underlying pays no dividends.
- No transaction costs are charged.
- No arbitrage opportunity exists.

Some of the previous points can be easily relaxed. Following researches have extended the Black-Scholes approach to allow for the parameters of the Brownian motion and the interest rate to be functions of time. Dividends can also be introduced without increasing the complexity of the result. In the last decades, papers considering the σ parameter of the Brownian motion as a random variable have been presented. However, for the purpose of this work, the standard Black-Scholes theory will be considered. All the mentioned extensions, but stochastic volatility, can be easily integrated in the results.

2.4.2 Deriving the Black-Scholes equation

The pricing for an option is conceptually based on the same no-arbitrage principle presented in Section 2.1. However, finding a risk-free portfolio is much more difficult. Let the value of the option be denoted by:

$$V\left(S, t, \sigma, \mu, K, T, r\right) \tag{2.4.1}$$

where:

- The variables S and t denote, respectively, the price of the underlying and the time
- μ and σ are parameters of the Brownian motion followed by the underlying
- K is the strike price, T the maturity and r the risk-free interest rate

From now on, parameters will be excluded from the notation: $V\left(S, t, \sigma, \mu, K, T, r\right) = V\left(S, t\right)$.

Let Π be the value of a portfolio created by buying a call option and short-selling a certain amount Δ of the underlying. Thus:

$$\Pi = V\left(S, t\right) - \Delta S \tag{2.4.2}$$

The change in value of the portfolio is:

$$d\Pi = dV - \Delta dS \tag{2.4.3}$$

As V is a function of S and t, it is possible to apply Ito's lemma:

$$dV = \left(\frac{\partial V}{\partial S}\mu S + \frac{\partial V}{\partial t} + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}\sigma^2 S^2\right) dt + \frac{\partial V}{\partial S}\sigma S \cdot dW$$

$$
\begin{aligned}
&= \frac{\partial V}{\partial t}dt + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}\sigma^2 S^2 dt + (\mu S dt + \sigma S \cdot dW)\frac{\partial V}{\partial S} \\
&= \frac{\partial V}{\partial t}dt + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}\sigma^2 S^2 dt + \frac{\partial V}{\partial S}dS
\end{aligned}
\tag{2.4.4}
$$

Coming back to the whole portfolio:

$$
d\Pi = dV - \Delta dS = \frac{\partial V}{\partial t}dt + \frac{1}{2}\frac{\partial^2 V}{\partial S^2}\sigma^2 S^2 dt + \left(\frac{\partial V}{\partial S} - \Delta\right) dS \tag{2.4.5}
$$

Now, the whole equation contains only one stochastic term: dS. If the quantity Δ is chosen such that:

$$
\Delta = \frac{\partial V}{\partial S} \tag{2.4.6}
$$

then, no uncertainty remains. As the option value continues to vary with respect to the underlying price, to perform this strategy it is required to continuously trade the underlying.

This is called delta-hedging.

A continuous delta-hedging is not feasible in practice; however, big companies which benefit of negligible transaction costs may get close to the theory.

If delta-hedging is performed, the portfolio Π becomes riskless. As per the no-arbitrage principle, the gain one can extract from it cannot differ from the riskless interest rate r.

$$
\begin{aligned}
d\Pi &= r\Pi dt \\
&= r\left(V - \Delta S\right) dt \\
&= r\left(V - \frac{\partial V}{\partial S} S\right) dt
\end{aligned}
\tag{2.4.7}
$$

By comparing Equations 2.4.5, corrected with delta-hedging strategy, and 2.4.7 and rearranging, the final Black-Scholes equation is found:

$$
\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV + rS\frac{\partial V}{\partial S} = 0 \tag{2.4.8}
$$

2.5 Pricing options: the Black-Scholes formulas

The Black-Scholes equation 2.4.8 is a Partial Derivative Equation (PDE) where the unknown function is the value V of the option. Remarkably, it does not depend on the type of option. This information is provided by the final condition of the PDE.

For a European call option, it would be:

$$
V(S,T) = \max\left(S(T) - K, 0\right) \tag{2.5.1}
$$

While, for a put option:

$$
V(S,T) = \max\left(K - S(T), 0\right) \tag{2.5.2}
$$

It was the last ingredient required to solve the PDE and get the formula to price the option. For European vanilla options a closed-form solution exists, while for exotic options numerical integration is usually the only viable way.

At least four alternative ways of solving the Black-Scholes equation have been proposed over the years. They can be found, for example, in [32].

Formulas for pricing call and put options are provided by the following lemmas.

Lemma 1 (Vanilla option price).
Under the Black-Scholes hypothesis, the price c of a European call option and the price p of a European put option are:

$$c = S_0 N(d_1) - Ke^{-rT} N(d_2) \tag{2.5.3}$$
$$p = Ke^{-rT} N(-d_2) - S_0 N(-d_1) \tag{2.5.4}$$

where:

$$d_1 = \frac{\ln(S_0/K) + \left(r + \sigma^2/2\right) T}{\sigma\sqrt{T}} \tag{2.5.5}$$
$$d_2 = \frac{\ln(S_0/K) + \left(r - \sigma^2/2\right) T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T} \tag{2.5.6}$$

S_0 is the price of the underlying at time $t = 0$, that is when the option is traded, and $N(x)$ is the cumulative probability distribution of a standard Gaussian variable.

2.5.1 Remarks on Black-Scholes formula

The most shocking feature of the Black-Scholes formula is probably its independence from the parameter which drives the trend of the underlying, μ. A deeper insight reveals that μ is wiped out by delta-hedging. As the portfolio used in the Black-Scholes argument is riskless, the value of the option logically does not depend from the trend of the underlying.

Despite appearing fairly complicated, the terms of the Black-Scholes formula have a straightforward interpretation.

- $N(d_2)$ is the probability of exercising the option if its expected return were the risk-free interest rate r.
- Ke^{-rT} is the actual value of the strike. As the strike price is paid only at maturity, it is worth less than if it were paid at the start of the contract. This is because, in the meanwhile, the owner of the option can put that money in the bank and receive the interests.
- $S_0 N(d_1)$ turns out to be exactly the actual value of the expected value of the underlying price, when the underlying price is higher than the strike.

Using the previous considerations, an interpretation of the formula is:

> The price of an option is the expected increase in value of the stock contingent to the exercise of the option minus the strike, weighted by the probability the strike is effectively paid, if expected return of the option were the risk-free interest rate. Both terms are discounted to get their present value.

Among all the parameters which appear in the Black-Scholes formula, S_0, K and T are established by the option contract. The risk-free interest rate r can be inferred from the interest rates the central banks apply or from riskless zero-cupon bonds.

The σ parameter of the geometric Brownian motion only remains an open point. The following section explains why.

2.5.2 Spread options: Margrabe's formula

Before leaving the Black-Scholes world, one last formula has to be presented. Derived by William Margrabe in 1978 [21], five years after the Black-Scholes formula was published, this equation results in the no-arbitrage pricing of a spread option which allow the exchange of an asset for another.

Equivalently, it leads to the pricing of a derivative whose payoff is:

$$P = \max\left(S_1\left(T\right) - S_2\left(T\right), 0\right) \tag{2.5.7}$$

Where S_1 and S_2 are the prices of the underlyings. The result is formalized in the following lemma.

Lemma 2 (Spread option price).
Under the hypothesis of the Black-Scholes world, the price V of an option to exchange two underlyings S_1 and S_2 in a time T from the signature of the contract is:

$$V = S_1\left(0\right) N\left(d_1\right) - S_2\left(0\right) N\left(d_2\right) \tag{2.5.8}$$

where:

$$d_1 = \frac{\ln\left(S_1\left(0\right)/S_2\left(0\right)\right) + \left(\sigma^2/2\right) T}{\sigma\sqrt{T}} \tag{2.5.9}$$

$$d_2 = d_1 - \sigma\sqrt{T} \tag{2.5.10}$$

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\sigma_1\sigma_2\rho} \tag{2.5.11}$$

σ_1 and σ_2 are the volatilities of the two underlyings, ρ their correlation and $N\left(x\right)$ is the cumulative probability distribution of a standard Gaussian variable.

Now nothing more prevents from tackling the problem of volatility.

2.6 The problem of volatility

The σ parameter of the Black-Scholes formula models the volatility of the underlying. Volatility is formally defined as the standard deviation of the annualized return. In plain English, the higher the volatility, the more bumpy is the time series of the stock prices. Thus, the higher the volatility, the more the future price is uncertain.

To better appreciate the concept, the following chart depicts three paths of geometric Brownian motions with different volatilities.

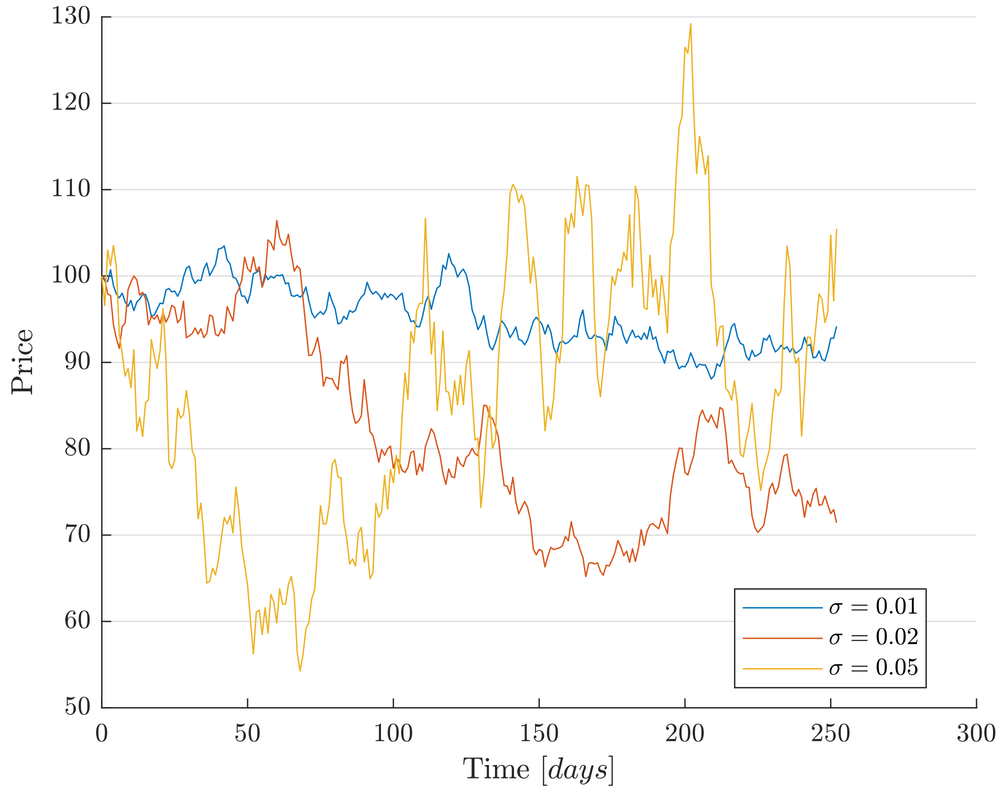


Figure 2.6.1: Geometric Brownian motion with different choices for σ.

In the Black-Scholes formula, when volatility appears, it is always multiplied by $\sqrt{T}$. To understand why this happens, one should recall Section 2.3.3. The evolution of the log-spot price $X = \log S$ is driven by a Wiener process. By definition, the standard deviation of a Wiener process scales as the square root of time. So, $\sigma\sqrt{T}$ can be read as the standard deviation of the log-spot price at time T.

Many different methods have been proposed over the years to estimate the volatility of an underlying. In the following sections, the two most common ones are reviewed. The first is based on the evolution of the stocks in some period in the past. It is thus called historical estimation. The second relies on the Black-Scholes formula to infer the volatility from the actual market price of options.

2.6.1 Historical estimation

The simplest approach is just to infer the standard deviation of the prices of the underlying from their past evolution. A common estimator for the actual standard deviation is the sample standard deviation, defined as:

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}\left(R_i - \bar{R}\right)^2} \tag{2.6.1}$$

where:

- R_i is the return at time step i out of n

- $\bar{R}$ is the average of all R_i

Usually daily prices are considered and, following the assumption returns are normally distributed, $\bar{R}$ is set to zero. Moreover, by taking advantage of Equation 2.3.13, the previous formula may become:

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n} \ln^2\left(\frac{S_i}{S_{i-1}}\right)} \tag{2.6.2}$$

Once the daily volatility s has been extracted, the estimation of the annual volatility $\hat{\sigma}$ can be derived from the properties of Brownian motion. It holds:

$$\begin{aligned} s &= \hat{\sigma}\sqrt{\Delta t} \\ \hat{\sigma} &= \frac{s}{\sqrt{\Delta t}} \end{aligned} \tag{2.6.3}$$

As $\hat{\sigma}$ is measured over years while s over days, Δt is the value of one day in terms of years. Conventionally, it is considered equal to 1/252, 252 being the number of trading days in one year. Why trading days? French and Roll [12] showed that the main source of volatility is the trading activity itself. The contribution to volatility given by the non-trading days is negligible.

2.6.2 Implied volatility

Despite being simple and easy to compute, historical volatility shows three main weaknesses:

- The past behaviour of an asset is not necessarily representative of its future evolution.
- The volatility is assumed to be constant. It is easy to understand that this is not true in the real world.
- The historical volatility fails in following fast changes in the market scenario.

Several studies have been published to support the previous points, but a chart may be enough to prove the previous statements. The Chicago Board Options Exchange's Volatility Index (VIX) provides a measure of the volatility of the options on the S&P 500 index, which is representative of the largest industrial companies of the United States. The following chart displays the time series of VIX from 1986 to 2017.

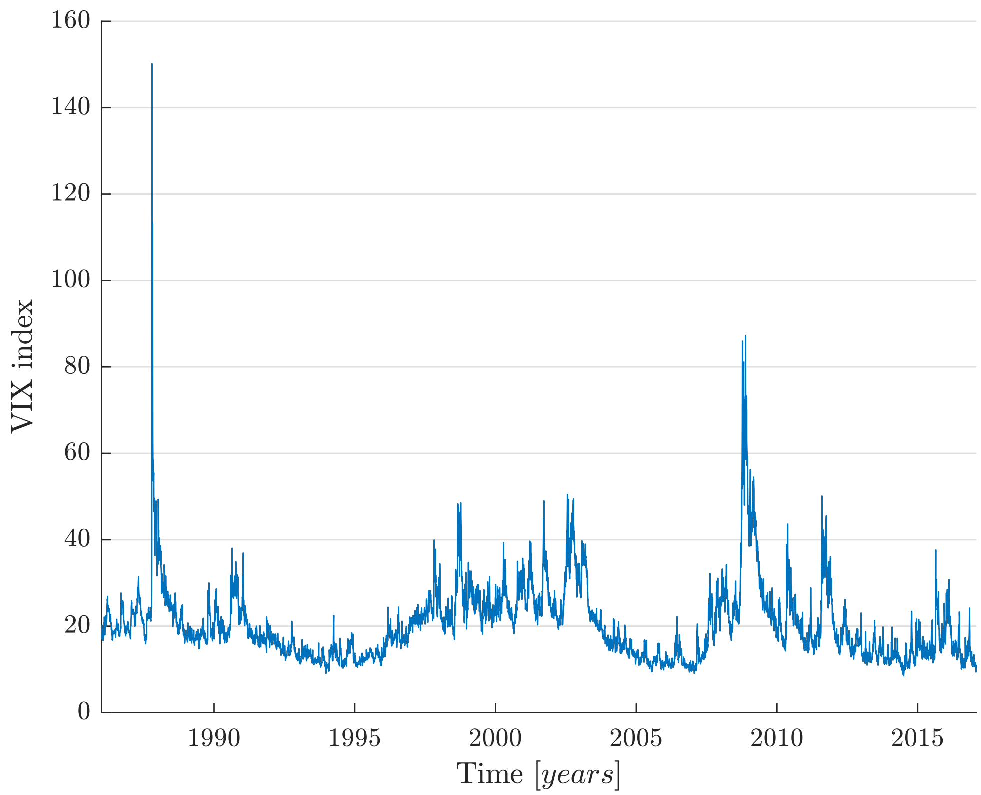


Figure 2.6.2: Chicago Board Options Exchange's Volatility Index (VIX) from 1986 to 2017.

The chart clearly shows that the volatility is not constant: this is particularly evident when great traumatic events happen - for example the market crashes in 1987 and the derivative crisis in 2008 - but is true at any time.

This is why traders usually prefer to derive volatility from the market prices. If the market price for a European option is known, it is possible to invert the Black-Scholes equation and solve for σ. No closed-form solution are available, but numerical methods can easily do the job.

The result is called implied volatility. As the implied volatility is derived from the market, it encodes the opinion the market has today on the uncertainty on the underlying in the future.

2.6.2.1 Implied volatility: a practical usage.

This approach is particularly useful whenever a company is required to write an option and sell it to another party. The question the company has to answer is: what is a fair price for the option? If all the parameters of the Black-Scholes formula were known, the answer would be pretty easy: the Black-Scholes price. But the volatility, especially the future volatility, is not usually known. If a market where similar options are traded exists, however, the seller can derive the implied volatility of traded options and use it to price his own derivatives. This way, he is sure that the price he is asking tails the market vision of the future.

A slightly more sophisticated version of this approach is used in Section 5.

2.6.3 State of the art

As soon as the Black-Scholes formula became public, the problem of estimating volatility was apparent. Historical volatility was the obvious solution and it was used from the very start [5]. At the same time, the possibility of inverting the equation to get to the implied volatility was detected [20]. With several improvements and modifications, these two streams dominated the scenes until stochastic volatility was introduced [31, 16].

Stochastic volatility models assume that the volatility is no longer a parameter, but a stochastic process itself. Stochastic volatility models was proven to be more effective in explaining the price of currency and bonds options than the Black-Scholes classical model.

Despite decades of literature on the topics, in the 2000s there still was debate on the accuracy of the models for volatility [4]. Incredibly enough, researches found a great deal of misunderstanding about the concept itself of volatility among the insiders - a famous experiment is reported in the paper "We Don't Quite Know What We are Talking About When We Talk About Volatility" [10].

Following the consolidated and overall accepted Black-Scholes theory of pricing, this work wants to propose a new way of deriving the volatility of the underlying. Despite being a general framework, the approach is particularly intended to be applied to electricity and gas markets.

Part II

A new approach to a classic problem

Chapter 3

Stochastic models and Lyapunov equation

The only unknown parameter in the Black-Scholes formula is volatility. Volatility always appear multiplied by a $\sqrt{T}$ factor, because of the model of the underlying. However, it has been noted that the standard geometric Brownian motion model does not fit the features of Energy commodity markets.

The basic idea underlying this and the next chapters is simple.

1. Find a more effective model of the underlying.
2. Reformulate it as a stochastic linear system.
3. Take advantage from the theory of dynamic systems to compute the variance of the state and then its standard deviation.
4. Plug the result into the Black-Scholes formula.

This chapter introduces the basic tools of the analysis of stochastic linear systems.

At the very beginning, white noise, the input of stochastic systems, is defined, and its relationship with the Wiener process is highlighted.

After that, stochastic linear systems are formally defined, and the formulas for the evolution of the mean and the variance of the state are derived. Particular attention is devoted to the variance of the state and to the equation which rules it: the matrix differential Lyapunov equation.

A selection of the models proposed in the literature is then reviewed and the one best suited to the peculiarities of Energy commodity markets is selected.

Finally, a simple procedure to translate the model into state equation is presented.

3.1 A technical aside: Wiener process and white noise

In this chapter, another interpretation of the Wiener process is introduced. A Wiener process W is defined as the integral of a continuous-time white noise u:

$$W(t) = \int_{-\infty}^{t} u(\tau)\, d\tau \tag{3.1.1}$$

Thus implying:

$$\frac{dW}{dt} = u(t) \tag{3.1.2}$$

A continuous-time white noise is defined by the following properties:

- $u(t)$ is normally distributed for each $t > 0$
- $E[u(t)] = 0, \ \forall t > 0$
- $Cov[u(t_1), u(t_2)] = \sigma^2 \delta (t_1 - t_2)$, where δ denotes the Dirac's delta distribution

In order to let the definition of white noise fit the stochastic systems theory, where inputs are usually vectors, sometimes the real number σ^2 is replaced by a square positive definite matrix S.

As the Dirac's delta distribution tends to infinity when its argument equals 0, the white noise does not have finite variance. It thus cannot be thought as a real-world process. However, it is widely used in the theory of stochastic linear systems and in many other fields.

3.2 Continuous-time stochastic linear systems

A continuous-time stochastic system is a dynamic system where the input is a random process.

Of particular interest are stochastic linear systems, as they admit closed-form solutions for both the variance and the mean of the state as functions of time.

When the output is not affected by noise, a continuous-time stochastic linear system fed by white noise may be described by the following stochastic differential equations.

$$\begin{cases} \frac{dx(t)}{dt} = Ax(t) + Bu(t) \\ y(t) = Cx(t) \end{cases} \tag{3.2.1}$$

where all the variables are matrices of suitable dimensions and the following assumptions hold:

- $u(t)$ is a continuous-time white noise with zero mean and covariance matrix

$$Cov[u(t_1), u(t_2)] = S\delta (t_1 - t_2) \tag{3.2.2}$$

- $u(t_1)$ and $x(t_2)$ are independent for each t_1 and t_2

To complete the description of the system, initial values for both the expected value and the variance of the state are required:

$$\bar{x}_0 := E[x(0)] \tag{3.2.3}$$

$$P_0 := Var[x(0)] \tag{3.2.4}$$

3.2.1 Expected value

Starting from equation 3.2.1, it is possible to derive the expected value of the state $x(t)$ for each time instant $t > 0$. The evolution of the state is ruled by the equation:

$$\frac{x(t+dt) - x(t)}{dt} = Ax(t) + Bu(t)$$
$$x(t+dt) = x(t) + Ax(t)\,dt + Bu(t)\,dt \tag{3.2.5}$$

By taking the expected value, one gets the following formula:

$$\begin{aligned} E[x(t+dt)] &= E[x(t)] + E[Ax(t)\,dt] + E[Bu(t)\,dt] \\ &= E[x(t)] + AE[x(t)]\,dt \\ \frac{E[x(t+dt)] - E[x(t)]}{dt} &= AE[x(t)] \end{aligned} \tag{3.2.6}$$

By taking the limit for $dt \to 0^+$ it is possible to get the differential equation which describes the evolution of the expected value of the state. It is worth noting that the following is a deterministic ordinary differential equation, as the expected value operator takes away the stochasticity:

$$\frac{d}{dt}\{E[x(t)]\} = AE[x(t)] \tag{3.2.7}$$

The equation can be easily integrated to get to the following formula:

$$\bar{x}(t) \coloneqq E[x(t)] = \bar{x}_0 e^{At} \tag{3.2.8}$$

Once the expected value of the state is known, the linearity of the expectation can be leveraged to get the expected value of the output:

$$E[y(t)] = E[Cx(t)] = C\bar{x}(t) = C\bar{x}_0 e^{At} \tag{3.2.9}$$

3.2.2 Variance: the Lyapunov equation

A closed-form solution for the variance of the state shall now be derived. The starting point is equation 3.2.5: however, the variance operator is now applied.

$$\begin{aligned} Var[x(t+dt)] &= Var[x(t) + Ax(t)\,dt] + Var[Bu(t)\,dt] \\ &= Var[(I + Adt)\,x(t)] + Var[Bu(t)\,dt] \\ &= (I + Adt)\,Var[x(t)]\,(I + Adt)^T + BVar[u(t)\,dt]\,B^T \\ &= (I + Adt)\,Var[x(t)]\left(I + A^T dt\right) + BVar[u(t)\,dt]\,B^T \end{aligned} \tag{3.2.10}$$

The term $u(t)\,dt$ represents the differential of a Wiener process. As shown in Chapter 2, the variance of the an infinitesimal increment of a Wiener process is Sdt. To make the following steps clearer, the following symbols are defined:

$$P(t) \coloneqq Var[x(t)] \tag{3.2.11}$$
$$Sdt \coloneqq Var[u(t)\,dt] \tag{3.2.12}$$

By performing computations, rearranging terms and dividing by dt, it is possible to write the infinitesimal increment of P as follows:

$$\frac{P\left(t+dt\right)-P\left(t\right)}{dt}=AP\left(t\right)+P\left(t\right)A^{T}+AP\left(t\right)A^{T}dt+BSB^{T} \tag{3.2.13}$$

As the time increment dt tends to zero, a differential equation in the matrix $P\left(t\right)$ is achieved:

$$\frac{dP\left(t\right)}{dt}=AP\left(t\right)+P\left(t\right)A^{T}+BSB^{T} \tag{3.2.14}$$

Once the solution $P\left(t\right)$ of the previous equation is found, it is immediate to compute the variance of the output $y\left(t\right)$.

$$Var\left[y\left(t\right)\right]=Var\left[Cx\left(t\right)\right]=CP\left(t\right)C^{T} \tag{3.2.15}$$

Equations like 3.2.14 are well-known in the system control literature.

Definition 2 (Matrix differential Lyapunov equation)**.**
An equation in the form

$$\frac{dP\left(t\right)}{dt}=AP\left(t\right)+P\left(t\right)A^{T}+Q \tag{3.2.16}$$

under the initial condition $P\left(0\right)=P_0=P_0^T$ *is known as the matrix differential Lyapunov equation [13].*

Because of its linearity, equation 3.2.14 has a unique solution. Moreover, as the system to study is linear and time-invariant, the instant at which the initial condition is evaluated can be set to $t_0=0$.

The solution of Equation 3.2.14 is given by the matrix version of the Lagrange formula [19, 13]:

$$P\left(t\right)=e^{At}P_0e^{A^Tt}+\int_0^t e^{A(t-z)}BSB^Te^{A^T(t-z)}dz \tag{3.2.17}$$

Once a model for the stochastic behaviour of the stock price is chosen and written as a stochastic linear model, it is therefore possible to solve the previous formula to get the value of the variance at each time instant. As the volatility of a stock is linked to its variance, it would be straightforward to derive a value to be plugged into the Black-Scholes formula.

3.3 Choice of the model

3.3.1 Single-factor models

The Black-Scholes formula itself assumes that the evolution of the underlying price is ruled by a Geometric Brownian Motion [3].

$$ds\left(t\right)=\mu s\left(t\right)dt+\sigma s\left(t\right)dW\left(t\right) \tag{3.3.1}$$

Such a model would be problematic to study from a dynamic system perspective, as it contains the product between the state $s\left(t\right)$ and the stochastic input $W\left(t\right)$. This feature makes the system non-linear and prevents the usage of the tools presented for linear systems.

However, it is possible to let the state be a function of the underlying price instead of the price itself to preserve linearity. If one defines the log-spot price:

$$x(t) = \ln(s(t)) \tag{3.3.2}$$

it has been shown in Section 2.3.3 that x follows a generalized Wiener process. Thus, it holds:

$$Var[x(t)] \propto t \tag{3.3.3}$$

thus implying that the standard deviation of the log-spot price is proportional to $\sqrt{t}$.

3.3.2 Mean reversion

The previous very simple model, however, does not keep into account the phenomenon of mean-reversion. The prices of energy commodities, as well as their log-spot prices, tend to follow a long-term trend: whenever they get away from it, for any reason, they tend to be pushed back to the trend within a short amount of time.

To better understand mean-reversion, let us consider the following example.

The price of electricity in the Northern Italy is slowly decreasing, due to the low price of oil and the ever-increasing efficiency of alternative energy sources. Suddenly, a very hot season hits the region. Usage of air conditioning devices makes the energy demand grow and, for the most basic law of economy, the price of energy rises. However, as soon as this happens, new actors, previously thwarted by higher production costs, enter the market. The offer grows while the temporary meteorological abnormality vanishes: the prices fall down again to their long-term run.

This situation is well presented by the following chart. It represents three years of daily average prices of the Italian Electricity market (PUN). Several seasons may be identified, showing a clear trend and sudden mean-reverting diversions from it.

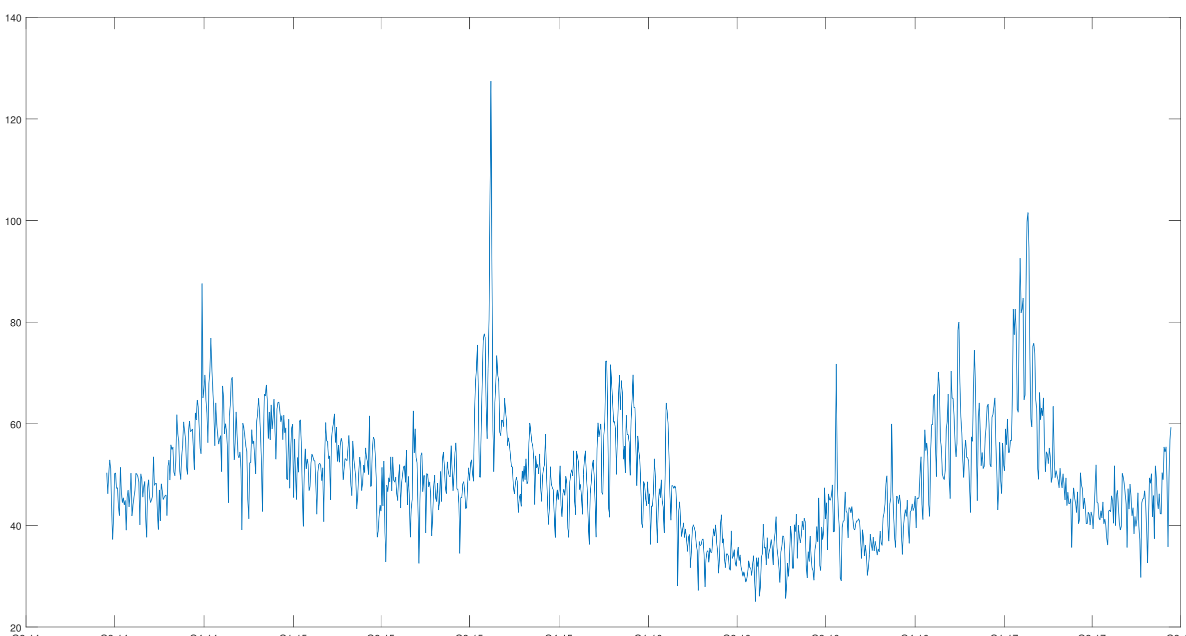


Figure 3.3.1: Three years of electric energy average daily prices.

The simplest mean-reverting model is the following:

$$\begin{cases} x(t) = \ln s(t) \\ dx(t) = \lambda(\mu - x(t))\, dt + \sigma dW \end{cases} \tag{3.3.4}$$

where:

- μ represents the long-term drift
- λ accounts for the strength of the reversion of $x(t)$ towards its long-term average. It is thus required $\lambda > 0$
- σ represents the volatility of the log-spot price
- $W(t)$ is a continuous-time Wiener process with zero mean: $W \sim N(0, t)$

It can be easily transformed into a stochastic linear system in form 3.2.1 by setting:

$$A = \left[-\lambda\right]$$

$$B = \left[\sigma\right] \tag{3.3.5}$$

$$C = \left[1\right] \tag{3.3.6}$$

The μ parameter can be set to 0, as it does not impact volatility.

The solution of the Lyapunov equation 3.2.14 is also straightforward. First, one should define

$$P_0 = p_0 \in \mathbb{R},\ p_0 > 0 \tag{3.3.7}$$

Now it is possible to solve:

$$\begin{aligned} P(t) &= e^{At} P_0 e^{A^T t} + \int_0^t e^{A(t-z)} BSB^T e^{A^T(t-z)} dz \\ &= e^{-\lambda t} p_0 e^{-\lambda t} + \int_0^t e^{-\lambda(t-z)} \sigma\sigma e^{-\lambda(t-z)} dz \\ &= p_0 e^{-2\lambda t} + \frac{\sigma^2}{2\lambda}\left(1 - e^{-2\lambda t}\right) \end{aligned} \tag{3.3.8}$$

As $C = 1$, the variance $P(t)$ of the state $x(t)$ is the same of the output $y(t)$.

For any feasible choice of the parameters, the variance $P(t)$ is going to start from its initial value p_0 and converge exponentially to its limit value:

$$\lim_{t \to +\infty} P(t) = \frac{\sigma^2}{2\lambda} \tag{3.3.9}$$

To illustrate the time evolution of $P(t)$, parameters are valued with the results presented in [2], where a one-factor model has been calibrated on the Italian Electricity market (PUN).

The following chart presents the time evolution of the log-spot price volatility $\sqrt{P(t)}$ with $\lambda = 57.62$, $\sigma = 1.82$, $\mu = -0.01$ and $p_0 = 0$.

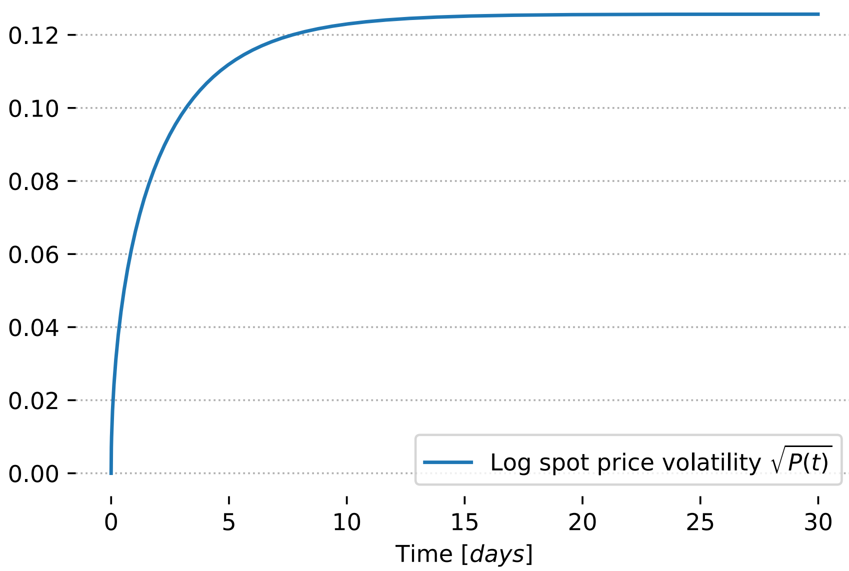


Figure 3.3.2: One factor model log-spot price volatility ($\sqrt{P(t)}$).

However, an asymptotically flat curve is not a good model of the volatility: in the real world, a non-increasing volatility means that the uncertainty on the log-spot prices - and thus the prices - does not increase with time. A realistic of volatility should include aspects of both geometric Brownian motion and mean-reverting models:

- An initial fast-growing volatility due to the diverting phenomena
- An asymptotic ever-increasing uncertainty

These arguments lead to the introduction of two-factors models, which take advantage of a set of two stochastic differential equations to provide an accurate description of both the long-term run and the short-term mean-reverting behaviour.

3.3.3 Two-factor models

The literature on stock price modelling provides several sophisticated models of the underlying price. Martin Barlow, Yuri Gusev and Manpo Lai [23] presented three different models:

- Log-spot price mean reverting to generalized Wiener process model (LMR-GW)
- Log-spot price mean-reverting to Ornstein-Uhlenback model (LMR-OU)
- Pilipovic model

All of the previous are two-factors models, meaning that the behaviour of the underlying price is represented by two stochastic differential equations.

The Pilipovic model is non-linear, as one of its equations contains the same term $\sigma S dW$ which appears in equation 3.3.1. This time, however, a logarithmic transformation cannot help in eliminating non-linearity.

Both the remaining models exploit one of their equations to model the short-term behaviour of the underlying, while the other takes care of the long-term evolution. The short-term component being the same, the only difference resides in the long-term equation.

LMR-GW employs a Generalized Wiener Process

$$dx(t) = \mu dt + \sigma dW(t) \tag{3.3.10}$$

while LMR-OU makes use of a Ornstein-Uhlenback process:

$$dx(t) = \lambda (L - x(t)) dt + \sigma dW(t) \tag{3.3.11}$$

The Ornstein-Uhlenback process introduces an additional parameter in the differential equation, namely in the term which accounts for the long-term drift. As shown in Section 2, the drift does not influence the pricing of the option, nor the volatility of the underlying. The principle of parsimony suggests not to introduce additional parameters in a model unless it is someway useful.

Moreover, the conclusion of the aforementioned paper highlights that LMR-OU model is the least effective model in explaining the behaviour of the spot prices, at least as far as the power market is concerned:

> "Based on the results of testing and parameter estimation, we reject the use of log-spot price mean-reverting to Ornstein-Uhlenback model for power markets. Its redundant complexity is not justified by any gains in modelling the market behaviour."

The previous arguments suggest to stick with the LMR-GW model, as far as volatility estimation is concerned.

3.4 Log-spot price Mean Reverting to Generalized Wiener process model (LMR-GW)

Let's consider a two-factor Log-spot price Mean Reverting to Generalized Wiener process model (LMR-GW). The evolution of the spot price of the underlying $s(t)$ is described by a standard mean-reverting model. The reversion, however, does not tail the drift of the spot price, but a different long-term behaviour, which is captured by another component. The whole model is described by the following set of stochastic differential equations:

$$\begin{cases} s(t) = e^{x_1(t)} \\ dx_1(t) = \lambda (x_2(t) - x_1(t)) dt + \sigma_1 dW_1(t) \\ dx_2(t) = \mu dt + \sigma_2 dW_2(t) \\ y(t) = x_1(t) \end{cases} \tag{3.4.1}$$

where:

- $s(t)$ is the spot price of the underlying
- $x_1(t) = \ln(s(t))$ and $x_2(t)$ are the short-term and the long-term components of the model
- $W_1(t)$ and $W_2(t)$ are continuous-time Wiener Processes with zero mean: $W_1(t) \sim N(0,t)$, $W_2(t) \sim N(0,t)$
- $y(t)$ is the observable output of the model
- $\lambda > 0$ accounts for the strength of the reversion of $x_1(t)$ towards $x_2(t)$
- σ_1 and σ_2 represent the different volatilities of the short-term and the long-term components
- μ represent the long-term drift of the underlying

The four parameters of the model (λ, μ, σ_1 and σ_2) can be estimated from the historical data through well-established techniques like Kalman filter [23] and Monte Carlo Markov Chain (MCMC) [2]. The parameters will thus be considered as given until Section 5, where a calibration procedure specific for the task of option pricing will be presented.

3.5 Transformation to a stochastic linear model

To take advantage from the theory of stochastic linear systems, an initial transformation of the model is required. By dividing the state equations of 3.4.1 by the time differential dt, one gets:

$$\begin{cases} \frac{dx_1(t)}{dt} = \lambda\left(x_2(t) - x_1(t)\right) + \sigma_1 \frac{dW_1(t)}{dt} \\ \frac{dx_2(t)}{dt} = \mu + \sigma_2 \frac{dW_2(t)}{dt} \\ y(t) = x_1(t) \end{cases} \tag{3.5.1}$$

The drift parameters μ does not play any role in the evolution of state variance: it can thus be set to 0. The system can now be rearranged in the standard form of stochastic linear systems by defining the following matrices:

$$A = \begin{bmatrix} -\lambda & \lambda \\ 0 & 0 \end{bmatrix} \tag{3.5.2}$$

$$B = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \tag{3.5.3}$$

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix} \tag{3.5.4}$$

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} \tag{3.5.5}$$

$$U(t) = \begin{bmatrix} \frac{dW_1(t)}{dt} \\ \frac{dW_2(t)}{dt} \end{bmatrix} \sim WN(0, I_2) \tag{3.5.6}$$

where WN denotes the White Noise and I_2 the second-order identity matrix. Equations 3.5.1 can now be written in the following equivalent form:

$$\begin{cases} \frac{dx(t)}{dt} = Ax(t) + Bu(t) \\ y(t) = Cx(t) \end{cases} \tag{3.5.7}$$

Chapter 4

A solution to the Lyapunov equation

The previous chapter showed that the variance of the state of a linear stochastic system obeys a differential Lyapunov equation. Now, it is time to perform the computation.

After a very brief summary on the origin of the Lyapunov equation, two approaches to its solution are presented.

The first one is based on the numerical approximation of a matrix exponential. It is easy, general, but relies on numerical methods to perform a couple of steps.

Then a closed-form solution is derived. It is necessarily model-dependent, but it bears the advantage of unveiling some pattern in the evolution of volatility.

A comparison between the two approaches closes the chapter. As expected, from both the viewpoints of accuracy and computational effort the analytical solution outperforms the numerical approach.

4.1 Variance in a stochastic linear system

As shown is section 3.2, given a stochastic linear system

$$\begin{cases} \frac{dx(t)}{dt} = Ax\left(t\right) + Bu\left(t\right) \\ y\left(t\right) = Cx\left(t\right) \end{cases} \tag{4.1.1}$$

where $U \sim WN\left(0, S\right)$ and S is a square, symmetric and positive definite matrix, it is known that the variance of the state $P\left(t\right) = Var\left[x\left(t\right)\right]$ is the solution of the matrix differential Lyapunov equation

$$\frac{dP\left(t\right)}{dt} = AP\left(t\right) + P\left(t\right)A^T + BSB^T \tag{4.1.2}$$

under the initial condition $P\left(0\right) = P_0$.

It is also known that the solution to 4.1.2 is given by the following formula:

$$P\left(t\right) = e^{At}P_0e^{A^Tt} + \int_0^t e^{A(t-z)}BSB^Te^{A^T(t-z)}dz \tag{4.1.3}$$

In the following paragraphs, solutions to the Lyapunov equation for model 3.5.1 are derived.

4.2 Numerical solution of differential Lyapunov equation

The numerical solution of the differential Lyapunov equation exploits the properties of matrix exponential and takes advantage of the following lemma, which is stated and proved in [6].

Lemma 3 (Exponential of triangular matrix).
Let A_{11}, A_{12} and A_{22} be matrices of suitable dimensions. Let

$$F = \begin{bmatrix} F_{11} & F_{12} \\ 0 & F_{22} \end{bmatrix} = \exp\left(\begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix} h \right) \tag{4.2.1}$$

Then the following equations hold:

$$F_{11} = e^{A_{11}h} \tag{4.2.2}$$

$$F_{22} = e^{A_{22}h} \tag{4.2.3}$$

$$F_{12} = \int_0^h e^{A_{11}(h-s)} A_{12} e^{A_{22}s} ds \tag{4.2.4}$$

Starting from the previous results, one can define a proper time-dependent $F(t)$ matrix as:

$$F(t) = \begin{bmatrix} F_{11}(t) & F_{12}(t) \\ 0 & F_{22}(t) \end{bmatrix} = \exp\left(\begin{bmatrix} A & BSB^T \\ 0 & -A^T \end{bmatrix} t \right) \tag{4.2.5}$$

to get:

$$F_{11}(t) = e^{At} \tag{4.2.6}$$

$$F_{22}(t) = e^{-A^T t} \tag{4.2.7}$$

$$F_{12}(t) = \int_0^t e^{A(t-z)} BSB^T e^{-A^T z} dz \tag{4.2.8}$$

The term F_{12} resembles the second item of the solution of the Lyapunov equation $\int_0^t e^{A(t-z)} BSB^T e^{A^T(t-z)} dz$. It is thus possible to manipulate the last exponential to get the rigorous equality:

$$\begin{aligned} F_{12}(t) &= \int_0^t e^{A(t-z)} BSB^T e^{-A^T z} dz \\ &= \int_0^t e^{A(t-z)} BSB^T e^{A^T(t-z)} e^{-A^T t} dz \\ &= \int_0^t e^{A(t-z)} BSB^T e^{A^T(t-z)} dz \cdot e^{-A^T t} \\ &= \int_0^t e^{A(t-z)} BSB^T e^{A^T(t-z)} dz \cdot F_{22}(t) \end{aligned} \tag{4.2.9}$$

The previous equation implies:

$$\int_0^t e^{A(t-z)} BSB^T e^{A^T(t-z)} dz = F_{12}(t) F_{22}^{-1}(t) \tag{4.2.10}$$

The whole solution of the Time-varying differential Lyapunov matrix equation can thus be expressed as follows - time dependency is omitted for readability:

$$P(t) = F_{11} P_0 F_{22}^{-1} + F_{12} F_{22}^{-1} = (F_{11} P_0 + F_{12}) F_{22}^{-1} \tag{4.2.11}$$

It is worth noting that, even if the final formulation contains a matrix inversion, no hypothesis on the A, B and S matrices should be introduced: $F_{22}(t) = e^{-A^T t}$ is always invertible.

This can be proven through the following theorem - stated and proved in [15] - and corollary.

Theorem 2 (Determinant of matrix exponential).
Let M be a square matrix. Then

$$\det\left(e^M\right) = e^{tr(M)} \tag{4.2.12}$$

Corollary 1 (Invertibility of matrix exponential).
Let M be a square matrix. Then e^M is invertible.

Proof. From Theorem 2 it holds:

$$\det\left(e^M\right) = e^{tr(M)} > 0 \tag{4.2.13}$$

So, the determinant of e^M is different from zero, which is a sufficient condition for e^M to be invertible. □

The whole procedure may be summarized by the following algorithm.

Algoritmo 4.1 General numerical algorithm to solve Time-varying differential Lyapunov matrix equation.

1. Take a linear stochastic system

$$\begin{cases} \frac{dx(t)}{dt} = Ax(t) + Bu(t) \\ y(t) = Cx(t) \end{cases} \tag{4.2.14}$$

   where $Var[x(0)] = P_0$.

2. Compute

$$F = \begin{bmatrix} F_{11}(t) & F_{12}(t) \\ 0 & F_{22}(t) \end{bmatrix} = \exp\left(\begin{bmatrix} A & BSB^T \\ 0 & -A^T \end{bmatrix} t\right) \tag{4.2.15}$$

3. Get the variance P of the state x as:

$$P(t) = \left(F_{11}(t) P_0 + F_{12}(t)\right) F_{22}^{-1}(t) \tag{4.2.16}$$

No assumptions on the matrices A, B and C of the linear stochastic system have been introduced. The procedure can thus be considered general-purpose. As long as the model for the underlying can be transformed into a linear stochastic system, Algorithm 4.1 can be applied.

4.3 Analytical solution of differential Lyapunov equation

Algorithm 4.1 provides a general-purpose solution of the Lyapunov equation. However, it heavily relies on the computation of a matrix exponential (step 2) and the inverse of a matrix (step 3). These operations must be performed by numerical routines which are implemented by all the most common scientific computation software and platforms - e.g MATLAB, R, Python and others.

Despite being heavily optimized, the numerical methods cannot match the execution speed of an analytical solution, where the only operation to be performed would be the evaluation of a function.

If the solution were evaluated once or even a small number of times, the difference would be negligible and could not be appreciated by a user. Unfortunately, model calibration presented in Section 5 requires the volatility $\sqrt{P_{11}(t)}$ to be plugged into an optimization routine. The most effective optimization algorithms are iterative, they thus require the target function to be evaluated several times.

Even worse, the optimization has to be performed on a large number of options.

By combining the two contributions, the number of evaluations of the formula in a common business day in an energy company can well overcome 10^5 - this happens in A2A, a major Italian energy provider, for instance. An even small difference in computational speed becomes a huge waste of time when multiplied by such a big factor - experiments are presented in Section 4.6 to confirm this common-sense consideration.

Hence the need for an analytical solution.

Moreover, an analytical solution may highlight some interesting properties of the result the numerical procedure hides.

In developing the computations to get to the analytical solution, generality is temporarily neglected: it is assumed that the model is a LMR-GW with a single underlying. Nonetheless, Chapter 7 provides a general, model-independent framework, where the only assumption is the diagonalizability of the dynamic matrix A.

4.3.1 Matrix exponential

The first issue to be addressed in the explicit computation of formula 4.1.3 is the matrix exponential e^{At}. A general formula to exactly compute the matrix exponential function is not available; however, under some hypothesis on the matrix A, the computation becomes straightforward.

For instance, if the matrix A is diagonal, its exponential is just the diagonal matrix which has the exponential of the elements of A on the main diagonal. Being the A matrix defined in 3.5.2 non-diagonal, a possible approach exploits the following lemma.

Lemma 4 (Exponential of diagonalizable matrix)**.**
Let $M \in \mathbb{M}_{\mathbb{R}}^{n\times n}$ be diagonalizable. That is, it exists a diagonal matrix $D \in \mathbb{M}_{\mathbb{R}}^{n\times n}$ and an invertible matrix $U \in \mathbb{M}_{\mathbb{R}}^{n\times n}$ such that $M = UDU^{-1}$. Therefore, $e^M = Ue^DU^{-1}$.

Matrix A defined in 3.5.2 is diagonalizable, as it is a second-order matrix with two distinct eigenvalues, $\varphi_1 = -\lambda$ and $\varphi_2 = 0$. Their eigenvectors can be easily found to be:

$$v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \tag{4.3.1}$$

D is a diagonal matrix made of the eigenvalues of A, while the U matrix is made of the eigenvectors of A. Therefore:

$$D = \begin{bmatrix} -\lambda & 0 \\ 0 & 0 \end{bmatrix} \tag{4.3.2}$$

$$U = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \tag{4.3.3}$$

Finally, the exponential of A can be obtained:

$$e^{At} = Ue^{Dt}U^{-1} = \begin{bmatrix} e^{-\lambda t} & 1-e^{-\lambda t} \\ 0 & 1 \end{bmatrix} \tag{4.3.4}$$

From the previous result all the other matrix exponentials mentioned in formula 4.1.3 can be derived through substitutions and the following property.

Lemma 5 (Exponential of transposed matrix)**.**
Let M be a square matrix. Then $e^{M^T} = \left(e^M\right)^T$.

4.3.2 First item of the formula

The initial state variance can be written, in the most general way, as a symmetric, positive definite matrix:

$$P_0 = \begin{bmatrix} p_1 & p_{12} \\ p_{12} & p_2 \end{bmatrix} \tag{4.3.5}$$

The first item in the solution of the Lyapunov equation can now be explicitly computed:

$$\begin{aligned} e^{At}P_0e^{A^Tt} &= \begin{bmatrix} e^{-\lambda t} & 1-e^{-\lambda t} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} p_1 & p_{12} \\ p_{12} & p_2 \end{bmatrix} \begin{bmatrix} e^{-\lambda t} & 1-e^{-\lambda t} \\ 0 & 1 \end{bmatrix}^T = \\ &= \begin{bmatrix} (p_1 - 2p_{12} + p_2)\, e^{-2\lambda t} + 2\,(p_{12} - p_2)\, e^{-\lambda t} + p_2 & (p_{12} - p_2)\, e^{-\lambda t} + p_2 \\ (p_{12} - p_2)\, e^{-\lambda t} + p_2 & p_2 \end{bmatrix} \end{aligned} \tag{4.3.6}$$

4.3.3 Second item of the formula

Before making any computation on the second element of the formula 4.1.3, it is handy to perform a simple substitution:

$$\tau = t - z \tag{4.3.7}$$

So that:

$$e^{A(t-z)}BSB^Te^{A^T(t-z)} = e^{A\tau}BSB^Te^{A^T\tau} \tag{4.3.8}$$

Matrix products can now be carried out:

$$\begin{aligned} e^{A\tau}BSB^Te^{A^T\tau} &= \begin{bmatrix} e^{-\lambda\tau} & 1-e^{-\lambda\tau} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix}^T \begin{bmatrix} e^{-\lambda\tau} & 1-e^{-\lambda\tau} \\ 0 & 1 \end{bmatrix}^T = \\ &= \begin{bmatrix} \sigma_1^2 e^{-2\lambda\tau} + \sigma_2^2\left(1-e^{-\lambda\tau}\right)^2 & \sigma_2^2\left(1-e^{-\lambda\tau}\right) \\ \sigma_2^2\left(1-e^{-\lambda\tau}\right) & \sigma_2^2 \end{bmatrix} \end{aligned} \tag{4.3.9}$$

4.3.4 Matrix integration

The following task is to solve the matrix integral:

$$\int_0^t e^{A(t-z)}BSB^Te^{A^T(t-z)}dz \tag{4.3.10}$$

By taking advantage of the substitution 4.3.7, the computation can be slightly simplified:

$$\int_0^t e^{A(t-z)} BSB^T e^{A^T(t-z)} dz = \int_0^t e^{A\tau} BSB^T e^{A^T\tau} d\tau \tag{4.3.11}$$

Therefore, component-wise integration is performed to get:

$$\begin{aligned}\int_0^t e^{A\tau} BSB^T e^{A^T\tau} d\tau &= \int_0^t \begin{bmatrix} \sigma_1^2 e^{-2\lambda\tau} + \sigma_2^2 \left(1-e^{-\lambda\tau}\right)^2 & \sigma_2^2 \left(1-e^{-\lambda\tau}\right) \\ \sigma_2^2 \left(1-e^{-\lambda\tau}\right) & \sigma_2^2 \end{bmatrix} dt = \\ &= \begin{bmatrix} -\frac{\sigma_1^2+\sigma_2^2}{2\lambda} e^{-2\lambda t} + 2\frac{\sigma_2^2}{\lambda} e^{-\lambda t} + \sigma_2^2 t + \frac{\sigma_1^2 - 3\sigma_2^2}{2\lambda} & \sigma_2^2 \left(\frac{e^{-\lambda t}}{\lambda} + t - \frac{1}{\lambda}\right) \\ \sigma_2^2 \left(\frac{e^{-\lambda t}}{\lambda} + t - \frac{1}{\lambda}\right) & \sigma_2^2 t \end{bmatrix}\end{aligned} \tag{4.3.12}$$

4.3.5 Final result

It is now possible to add formulas 4.3.6 and 4.3.12 to get to the final formulation of $P(t)$.

$$P(t) = \begin{bmatrix} P_{11}(t) & P_{12}(t) \\ P_{21}(t) & P_{22}(t) \end{bmatrix} \tag{4.3.13}$$

$$\begin{aligned} P_{11}(t) &= \left(p_1 - 2p_{12} + p_2 - \frac{\sigma_1^2 + \sigma_2^2}{2\lambda}\right) e^{-2\lambda t} + 2\left(p_{12} - p_2 + \frac{\sigma_2^2}{\lambda}\right) e^{-\lambda t} + \sigma_2^2 t + \frac{\sigma_1^2 - 3\sigma_2^2}{2\lambda} + p_2 \\ P_{12}(t) = P_{21}(t) &= \left(p_{12} - p_2 + \frac{\sigma_2^2}{\lambda}\right) e^{-\lambda t} + \sigma_2^2 t + p_2 - \frac{\sigma_2^2}{\lambda} \\ P_{22}(t) &= \sigma_2^2 t + p_2 \end{aligned} \tag{4.3.14}$$

4.4 Remark: long-term and short-term volatility

It is worth noting that, as time t tends to infinity, the exponential terms $e^{-2\lambda t}$ and $e^{-\lambda t}$ vanish, letting all the elements in the covariance matrix grow linearly, like $\sigma_2^2 t$. This is expected, as the long-term equation represent a standard Brownian motion model, whose variance is known to grow linearly with time. Volatilities, which are by definition standard deviations rather than variances, are thus expected to grow as $\sqrt{t}$.

Moreover,

$$\lim_{t\to+\infty} P_{11}(t) - P_{22}(t) = \frac{\sigma_1^2 - 3\sigma_2^2}{2\lambda} \tag{4.4.1}$$

As the long-term process is supposed to vary less fast than the short-term one, it is not wrong to suppose $\sigma_1^2 \gg \sigma_2^2$, this meaning that the uncertainty on the spot prices will always be greater than the on the long-term run.

In the very first time instant, the gap between the two is larger, due to the effect of the fast-vanishing exponential components. The same effect has already been detected by Eduardo Schwartz and James E. Smith [29] while applying slightly different models to study the evolution of oil prices.

The difference between short-term and long-term components of the variance is well highlighted when the time evolution of the two is plotted on the same chart. To translate the evidence provided by equation 4.3.14 in a well-defined function it is required to fix the value of the model parameters. Both the works of Barlow [23] and Marziali and coworkers [2] contain model calibration which provide realistic values. In the latter work, three years of Electricity market data were fed into a MCMC Bayesian estimation algorithm to find the probability distribution of the parameters. The following table summarizes the mean values of the parameters.

Parameter	MCMC mean
λ	207.5593
μ	0.0346
σ_1	1.4629
σ_2	0.4494

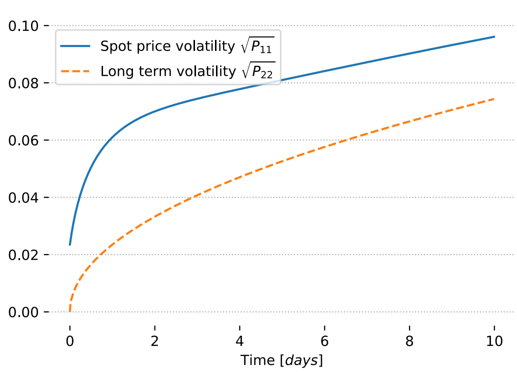
Table 4.1: LMR-GW model parameters

As far as the initialization of the P matrix is concerned, both works agree in choosing the matrix P_0 as:

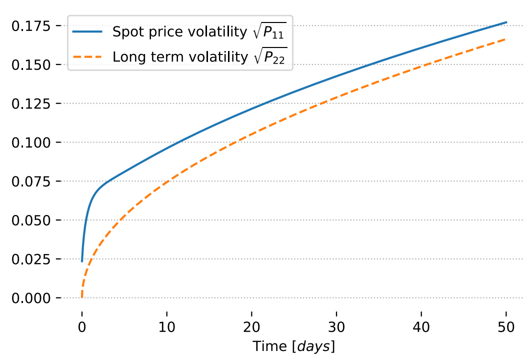
$$P_0 = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} = \begin{bmatrix} \sigma_2^2\delta & 0 \\ 0 & 0 \end{bmatrix} \tag{4.4.2}$$

where $\delta = 1/365$ days is the time step of the price series.

Among the four elements of the $P(t)$ variance matrix, the most interesting is $P_{11}(t)$. It represents the variance of the log-spot prices, which is by definition the square root of the volatility the Black-Scholes formula requires. It could be of some interest to show how it behaves with respect to the $P_{22}(t)$ component, which is the variance of the long-term run. To preserve dimensional consistency, square root is taken on both $P_{11}(t)$ and $P_{22}(t)$ to obtain volatilities. The following charts represent the evolution of $\sqrt{P_{11}(t)}$ and $\sqrt{P_{22}(t)}$ with different time horizons.



(a) Ten-days evolution.



(b) Fifty-days evolution.

Figure 4.4.1: Volatility over time.

The charts in Figure 4.4.1 show how the influence of the short-term volatility is limited to the very first days, mainly because of the magnitude of the λ parameter. In the real world, it means that the mean-reversion of the electricity market is very strong. Whenever the price goes away from its long-term run, it does not stay there for a long time. After the first few time steps, the difference between the long-term and the short-term components become constant and more and more negligible with respect to both values, as the theory predicts.

4.5 Proof of correctness

The simplest way to proof the correctness of the results provided by formula 4.3.14 is plugging the $P(t)$ matrix into the Lyapunov equation 3.2.14 and verifying that it is an identity.

The time derivative of $P(t)$ is:

$$\frac{dP(t)}{dt} = \begin{bmatrix} -2\lambda\left(p_1 - 2p_{12} + p_2 - \frac{\sigma_1^2+\sigma_2^2}{2\lambda}\right)e^{-2\lambda t} - 2\lambda\left(p_{12} - p_2 + \frac{\sigma_2^2}{\lambda}\right)e^{-\lambda t} + \sigma_2^2 & -\lambda\left(p_{12} - p_2 + \frac{\sigma_2^2}{\lambda}\right)e^{-\lambda t} + \sigma_2^2 \\ -\lambda\left(p_{12} - p_2 + \frac{\sigma_2^2}{\lambda}\right)e^{-\lambda t} + \sigma_2^2 & \sigma_2^2 \end{bmatrix} \tag{4.5.1}$$

The second member of the Lyapunov equation 3.2.14 can be computed as follows

$$\begin{aligned} AP(t) + P(t)A^T + BSB^T &= AP(t) + (AP(t))^T + BI_2B^T \\ &= AP(t) + (AP(t))^T + BB^T \\ &= \begin{bmatrix} -\lambda & \lambda \\ 0 & 0 \end{bmatrix}\begin{bmatrix} P_{11}(t) & P_{12}(t) \\ P_{12}(t) & P_{22}(t) \end{bmatrix} + \left(\begin{bmatrix} -\lambda & \lambda \\ 0 & 0 \end{bmatrix}\begin{bmatrix} P_{11}(t) & P_{12}(t) \\ P_{12}(t) & P_{22}(t) \end{bmatrix}\right)^T + \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix} \\ &= \begin{bmatrix} -2\lambda P_{11}(t) + 2\lambda P_{12}(t) + \sigma_1^2 & -\lambda P_{12}(t) + \lambda P_{22}(t) \\ -\lambda P_{12}(t) + \lambda P_{22}(t) & \sigma_2^2 \end{bmatrix} \end{aligned}$$

By introducing the elements of the matrix $P(t)$ found in the previous paragraph, one can obtain the final expression

$$\begin{aligned} &AP(t) + P(t)A^T + BSB^T = \\ &= \begin{bmatrix} -2\lambda\left(p_1 - 2p_{12} + p_2 - \frac{\sigma_1^2+\sigma_2^2}{2\lambda}\right)e^{-2\lambda t} - 2\lambda\left(p_{12} - p_2 + \frac{\sigma_2^2}{\lambda}\right)e^{-\lambda t} + \sigma_2^2 & -\lambda\left(p_{12} - p_2 + \frac{\sigma_2^2}{\lambda}\right)e^{-\lambda t} + \sigma_2^2 \\ -\lambda\left(p_{12} - p_2 + \frac{\sigma_2^2}{\lambda}\right)e^{-\lambda t} + \sigma_2^2 & \sigma_2^2 \end{bmatrix} \end{aligned} \tag{4.5.2}$$

As expressions 4.5.1 and 4.5.2 coincide, the equation is verified by the proposed solution.

4.6 Numerical and analytical solution: a comparison

Two approaches have been derived to get the time-dependent expression of the variance of the log-spot price. On the one hand, the numerical method is more general, as it does not rely on the model of the underlying, as long as it can be expressed by a linear stochastic system. On the other hand, the analytical solution provides deeper insight and, at least on paper, promises higher accuracy and lower computational effort. Moreover, a new approach to the solution of the Lyapunov equation (see Section 7) may make the method model-independent.

In order to inspect pros and cons of the two approaches, a comparison is now performed over the following points:

- Accuracy, defined as the relative error of the numerical solution to the analytical one
- Computational time, defined as the time required by an implementation of the approach to produce results

4.6.1 Accuracy

The proof in Section 4.5 ensures that, given the model of the underlying, the result of the analytical solution is correct. When mathematical formulas translate to software routines, the only limit is thus the arithmetic precision of the computer. This limitation is merely technical and unavoidable: thus it will not be taken into consideration.

For what concerns the numerical solution, however, the correctness cannot be taken for granted.

The solution relies on two numerical computations: a matrix exponential and a matrix inversion. Although both of them are very common problems and literature has provided over the years several robust and optimized algorithms, research is still active in the field. For example, most of the scientific computing software, including MATLAB and Python, compute the matrix exponential with an algorithm first described in 2009 [1].

Nonetheless, some cases are known where the matrix exponentiation $\exp(M)$ may cause troubles.

The numerical procedure currently used by most of the scientific computing software and described in [1] relies on the Padé approximation, which may suffer when the eigenvalues of M are widely spread [24] or M is highly ill-conditioned.

To avoid digging too deeper into numerical details which are not within the scope of this work, it may worth noting that several tests run on LMR-GW models with λ ranging from an unrealistic 10^{-5} to an even more unrealistic 10^{5} show that the difference between the analytical and the numerical solution is negligible. The order of magnitude of the absolute value of the difference between the two results never overcomes 10^{-15} when double precision floating point variables are employed.

The following chart represents a standard situation: the two curves perfectly overlap.

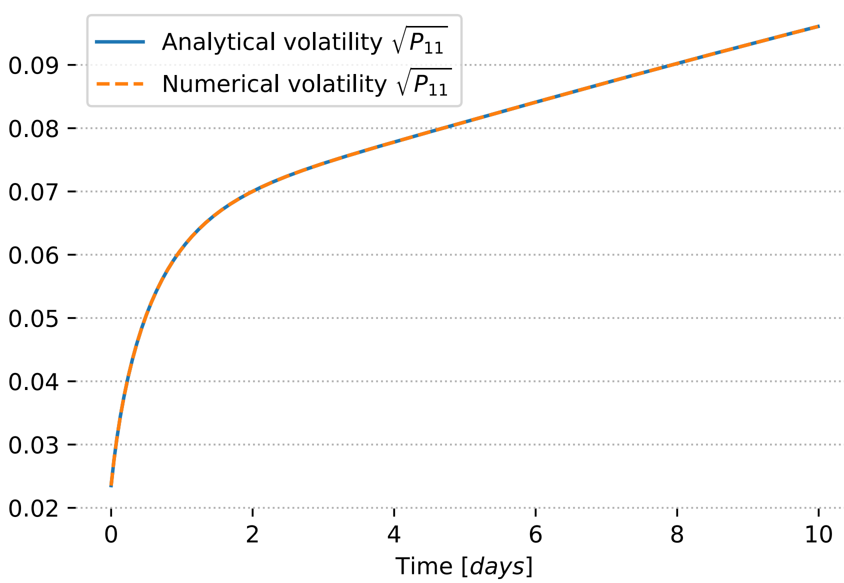


Figure 4.6.1: Volatility $\sqrt{P_{11}(t)}$ computed both with the numerical and the analytical approach.

However, there is another situation where a numerical matrix exponentiation may run into troubles.

The obvious case when M is not stable and its elements are too large. Whenever it happens, an overflow may arise.

When the numerical approach described in Section 4.2 is applied to an LMR-GW model, the stability of A matrix does no longer guarantee any bound on the elements of F, which was defined as:

$$F(t) = \begin{bmatrix} F_{11}(t) & F_{12}(t) \\ 0 & F_{22}(t) \end{bmatrix} = \exp\left(\begin{bmatrix} A & BSB^T \\ 0 & -A^T \end{bmatrix} t \right) \tag{4.6.1}$$

Indeed, if one considers a matrix A as defined by the LMR-GW model (see Equation 3.5.2):

$$A = \begin{bmatrix} -\lambda & \lambda \\ 0 & 0 \end{bmatrix}, \quad \lambda > 0 \tag{4.6.2}$$

the matrix $-A^T$ turns out to be unstable, as it becomes a lower triangular matrix with eigenvalues λ and 0.

$$-A^T = \begin{bmatrix} \lambda & 0 \\ -\lambda & 0 \end{bmatrix}, \quad \lambda > 0 \tag{4.6.3}$$

If λ is quite large and t grows, some element of F may become huge quite fast.

Using either linear algebra computations or the datapoints in the previous charts, it is possible to show that:

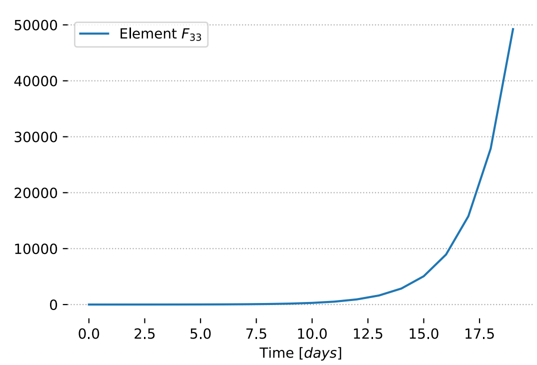
$$F_{33}(t) \propto e^{\lambda t} \tag{4.6.4}$$

where $F_{33}(t)$ denotes the entry $(3,3)$ of the time-varying matrix $F(t)$.

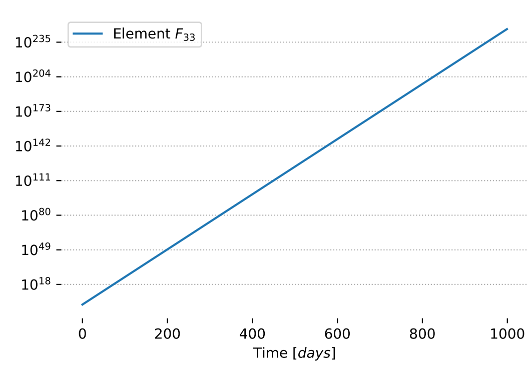
By convention, the unit of λ is the inverse of years. Given the maximum number a double-precision floating point variable is able to represent $N_{max} = 2^{1023} \approx 9 \cdot 10^{307}$, it is possible to invert the equation to get the time t_{max} when overflow arises.

$$\begin{aligned} e^{\lambda t_{max}} &= N_{max} \\ t_{max} &= \frac{\ln N_{max}}{\lambda} \end{aligned} \tag{4.6.5}$$

Unfortunately, as highlighted in Section 4.4, λ calibrated on the PUN electricity market is $\lambda = 207.56$. The following charts represent the element F_{33} as a function of the time t - the other parameters are the same as in Section 4.4.



(a) Twenty days evolution, linear scale.



(b) One thousand days evolution, semi-log scale.

Figure 4.6.2: Element F_{33} over time.

Keeping the same value of λ as in the example, it is possible to compute

$$t_{max} = \frac{\ln N_{max}}{\lambda} \approx 3.4183 \tag{4.6.6}$$

The numerical approach can thus handle time ranges which are not longer than, roughly, three and a half years.

This limit may become a big handicap when market calibration over a long time horizon is performed. As Section 5 explains in full detail, calibration takes advantage of the market prices of options to tune a model which is then used to price other non-traded options. The more the features of trading options are close to the ones of the options to price, the more the model is accurate. If the target options had maturity longer than t_{max}, the numerical approach could not be used in market calibration.

Moreover, a limit dependent on λ, one of the parameters to be tuned, may lead to error and unexpected behaviour of the software routines which automatically take care of the model tuning.

4.6.2 Computational effort

While for the accuracy the analytical solution was considered the touchstone, computational effort requires more caution. On paper, matrix exponentiation and inversion are very time-consuming algorithms. However, the size of the matrix they are applied on is limited by the problem formulation. As far as an LMR-GW model with a single underlying is considered, the matrix to exponentiate is always four by four. For the same reason, the F_{22} matrix to invert is always two by two. The theoretical asymptotic computational complexity of the algorithms does not provide the full picture. What really matters are the performances of the algorithms on small, simple matrices.

However, calibration requires the variance function to be evaluated several times and in a standard business day calibration should be run against thousands of different assets: even small differences in the execution time are hugely amplified.

This is the reason why tests were run to verify the execution time of the two methods.

Both the analytical and numerical approaches were implemented using the Python language, that has been chosen because of its flexibility, its steep learning curve and the availability of a great number of scientific libraries. Moreover, differently from Matlab or other computational environments, it is free and open source. The procedures required by the numerical solution are implemented by the Scipy package:

- Matrix exponential is carried out by the `expm` function of the `scipy.linalg` module. According to the official documentation [8], this function employs the Padé approximant, improved with scaling and squaring methods [1].
- Matrix inversion is implemented by the `inv` function of the `scipy.linalg` module through the Gaussian elimination method.

Both of the previous are state-of-the-art algorithms in their fields and they are quite standard throughout most of the scientific computing environments.

The most relevant features of the machine used for the tests are listed in the following table.

Processor	Intel Core i5 3340M
RAM	16 GB
Clock	2.7 GHz
Cores	2
Threads	4

Table 4.2: Features of the test machine.

One to $3 \cdot 10^4$ evaluations of both the analytical and the numerical solutions are run on different time instants.

Each run is repeated 10 times. The lower and the higher timings are discarded, from the eight remaining the median is taken as the most significant value. Median is preferred to mean as a statistical indicator because it is more robust to outliers - which may always arise, in particular when a small number of evaluation is considered. Both wall clock time and CPU time are traced.

- Wall clock time is captured through the `perf_counter` function of the standard `time` Python module
- Cpu working time is traced by the `process_time` function of the same module

The following table summarizes the results:

Evaluations	**Analytical - CPU**	**Analytical - wall**	**Numerical - CPU**	**Numerical - wall**
1	0.000000	0.000047	0.000000	0.001238
10	0.000000	0.000256	0.011625	0.012116
100	0.0025	0.0026	0.0938	0.0985
1000	0.0253	0.0259	0.9950	0.9956
3000	0.0731	0.0738	3.0547	3.0751
5000	0.1225	0.1239	5.0781	5.0849
10000	0.2813	0.2885	10.5313	10.5574
20000	0.4922	0.4958	20.7578	20.8032
30000	0.7344	0.7394	31.4531	31.4745

Table 4.3: Test results. All items are in seconds. "wall" refers to the wall clock time, while "CPU" identifies the CPU working time.

Since the simulations were run minimizing the number of concurrent jobs other than the program under test, the wall clock time and CPU time do not actually differ significantly. As the official Python documentation states, `perf_counter` provides an higher resolution than `process_time`. This is the reason why there are some zeros in the columns of CPU timings. With a small number of evaluations the program runs too fast for its duration to be appreciated by the counter.

A very first inspection reveals how the analytical solution is consistently faster than the numerical one. To better appreciate the difference, speedups S_n are computed for each number of evaluations n, according to the formula:

$$S_n = \frac{Numerical\ solution\ time}{Analytical\ solution\ time} \tag{4.6.7}$$

Results are provided by the following table.

Evaluations	Speedup - wall	Speedup - CPU
1	26.34	N/A
10	47.33	N/A
100	37.50	38.17
1000	39.41	38.44
3000	41.77	41.64
5000	41.45	41.04
10000	37.44	36.59
20000	42.17	41.96
30000	42.83	42.57

Table 4.4: Speedups. N/A denotes a not available entry, as the CPU time could not be captured.

Speedups are consistently over one order of magnitude and are almost constant for large enough amounts of evaluations. As the number of evaluations n increases, the single evaluation time of both the analytical solution T_A and the numerical approach T_N is multiplied by a factor n. The total time required by the analytical solution thus becomes:

$$Analytical\ solution\ time = nT_A \tag{4.6.8}$$

and the time taken by the numerical formula:

$$Numerical\ solution\ time = nT_N \tag{4.6.9}$$

The speedup can thus be expressed by the ratio:

$$S_n = \frac{Numerical\ solution\ time}{Analytical\ solution\ time} = \frac{nT_N}{nT_A} = \frac{T_N}{T_A} \tag{4.6.10}$$

Speedup is thus only explained by the intrinsic complexity of the procedures. The phenomenon is highlighted by plotting the computation times side by side.

The linear scale chart shows that both approaches scale linearly with the evaluation, but slopes are different - as Formulas 4.6.8 and 4.6.9 suggest. If a log scale in used for the vertical axes,

$$\log\left(Analytical\ solution\ time\right) = \log\left(nT_A\right) = \log n + \log T_A \tag{4.6.11}$$

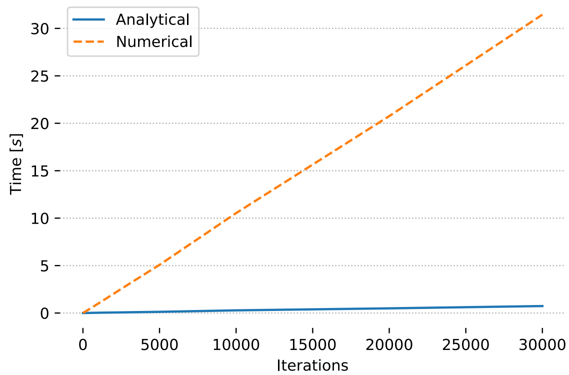
and

$$\log\left(Numerical\ solution\ time\right) = \log\left(nT_N\right) = \log n + \log T_N \tag{4.6.12}$$

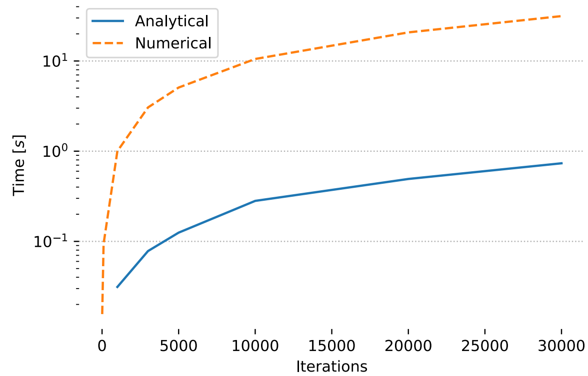
The difference between the two curves is thus constant and equal to the logarithm of the speedup ratio:

$$\log\left(Numerical\ solution\ time\right) - \log\left(Analytical\ solution\ time\right) = \log\frac{T_N}{T_A} \tag{4.6.13}$$

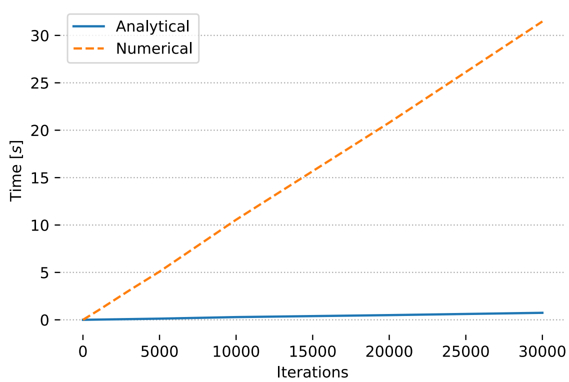
The charts are presented in the following figure.



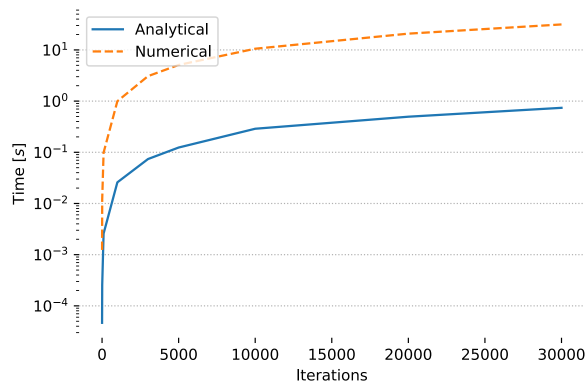
(a) CPU time: linear scale.



(b) CPU time: semi-log scale.



(c) Wall clock time: linear scale.



(d) Wall clock time: semi-log scale.

Figure 4.6.3: CPU and wall clock time against number of evaluations.

Chapter 5

Model calibration and option pricing

In Chapter 3 a mathematical model for the underlying has been developed. In Chapter 4 the variance of the underlying price as a function of time has been derived from that model. However, that function depends on the parameters of the model. Until now, they have been taken as given: it's time to show how they can be derived.

Before doing that, a very short summary of the Black-Scholes framework for option pricing is provided. Focus is put on how the results of the previous chapters may fit into it.

Focus then moves on model calibration. Tuning a model for the underlying is not different from finding the volatility to plug into the Black-Scholes formula. Two approaches can be followed: one wants the parameters to be inferred from the history of the underlying, the other from market prices of options.

Both the approaches are reviewed and implemented. Results are then tested on a case-study. Several European vanilla options of German electricity market are priced with both methods. Being those traded in regular exchanges, it is possible to assert to what extent the prices produced by the different approaches match the market.

In the end, another application of vanilla options pricing is presented: complex derivative pricing through Monte Carlo simulation.

5.1 From Black-Scholes theory: pricing European options

Let the assumptions of the Black-Scholes world, introduced in Chapter 2, hold and let all the notation defined in that chapter be valid. Lemma 1 states that the price of a European call option c and of a put option p is given by:

$$\begin{aligned} c &= S_0 N(d_1) - K e^{-rT} N(d_2) \\ p &= K e^{-rT} N(-d_2) - S_0 N(-d_1) \end{aligned} \tag{5.1.1}$$

where:

$$\begin{aligned} d_1 &= \frac{\ln(S_0/K) + \left(r + \sigma^2/2\right) T}{\sigma\sqrt{T}} \\ d_2 &= \frac{\ln(S_0/K) + \left(r - \sigma^2/2\right) T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T} \end{aligned} \tag{5.1.2}$$

It has also been shown that the term $\sigma\sqrt{T}$ is nothing but the standard deviation of the log-spot prices at time T. If a different model is used, such that the variance of underlying at time t is given by a new function $p(t)$, the term d_1 and d_2 can be written as follows:

$$
\begin{aligned}
d_1 &= \frac{\ln(S_0/K) + rT + \frac{1}{2}p(T)}{\sqrt{p(T)}} \\
d_2 &= \frac{\ln(S_0/K) + rT - \frac{1}{2}p(T)}{\sqrt{p(T)}} = d_1 - \sqrt{p(T)}
\end{aligned} \tag{5.1.3}
$$

In Chapter 4, an expression for the variance $P(t)$ of the states of a 2-factor LMR-GW model has been derived. However, the previous equation requires only the scalar-valued function $p(t)$. Here, the output transformation of the linear stochastic system comes into play. The output of the system, that is the log-spot price of the underlying, was defined in 3.5 as:

$$
y(t) = Cx(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = x_1(t) \tag{5.1.4}
$$

The variance of the log-spot price can now be found through the formula

$$
Var[y(t)] = CVar[x(t)]C^T = CP(t)C^T \tag{5.1.5}
$$

to be equal to the element $p_{11}(t)$ of the matrix $P(t)$. From now on, the variance of the log-spot prices will be denoted by:

$$
p(t) = p_{11}(t) \tag{5.1.6}
$$

With the results achieved for the 2-factor model finally linked to the Black-Scholes theory, it is time to tackle the problem of model calibration and parameter fitting.

5.2 Historical calibration

A first answer to the calibration problem comes from the past behaviour of the underlying. Electricity markets are still relatively young - German market opened in 2000, Italian market in 2004 - but several years of data are available. For oil or gas, markets are way more mature. One can thus fit the parameters to make the model explain as well as possible the past evolution of prices.

The problem of calibrating multifactor stochastic models against historical time series is not new in the literature. It has been tackled, for example, by Barlow, Gusev and Lai in [23]. They used a maximum likelihood estimation on the residual of a standard Kalman filter. The Kalman filter is required to estimate the evolution of the states of the stochastic model of the underlying. Over previous attempts, this method has the great advantage of relying only on the prices. All the information encoded in the model comes from the market and no other sources can introduce noise. Results on simulated data presented in the paper are very encouraging.

The same approach is analysed in [2] and compared with a Bayesian estimation. In each case study, parameters tuned through the Kalman filter are close to the Bayesian mean and median and compatible with the Bayesian distribution. The major enhancement of the Bayesian estimation is providing the full probability distribution of the parameters.

However, the Bayesian estimation relies on Monte Carlo Markov Chain (MCMC) methods, which are very demanding in terms of computational load. As far as this work is concerned, the need to generate fast result for test cases forces to opt for the Kalman filter.

5.3 Market calibration

Historical model calibration can be a good choice for forecasting. But, as far as pricing is concerned, it may be not the optimal solution. The same argument which led to the definition of implied volatility holds: there is no guarantee that the historical evolution of the underlying reflects the its behaviour in the future. More precisely, there is no guarantee that the historical behaviour tails the opinion of the market about the future evolution of the underlying. More precisely, the opinion the market has today about the future evolution of the volatility of the underlying.

This is why market calibration is introduced. The optimal value for model parameters are assumed to be the ones which better explain the price of the options on the underlying. This goal can be achieved by minimizing a proper loss function.

Let P_i, $i = 1, ..., n$ be n European vanilla options traded on a certain market. Let also $\hat{P}_i$ be the price of the i-th option given by the Black-Scholes formula 5.1.1. The loss function can be defined as follows:

$$L = \sum_{i=1}^{n} \left(P_i - \hat{P}_i \right)^2 \tag{5.3.1}$$

So that the set of optimal parameters

$$\theta = \{\lambda, \sigma_1, \sigma_2\} \tag{5.3.2}$$

can be found by computing

$$\min_{\theta} L\left(\theta\right) \tag{5.3.3}$$

Since both $L(\theta)$ and its gradient are highly non-linear function, numerical methods are required to perform the optimization.

Once the model has been calibrated, it may be used to price options which are not on the market. This way, the stochastic model of the underlying and the Black-Scholes formula act as a complex interpolation tool. The final goal is to provide traders with a price in line with the market.

5.4 Historical vs Market calibration: a comparison

The basic task a good calibration has to perform is making the model capable of producing prices in line with the market. The best measure to perform a comparison between historical and market calibration method is thus the difference between the price produced by tuned models and the actual market price.

A test case has been set up in order to perform a comparison between the two calibration approaches. The first choice to take is about the market. The range is limited to the markets whose data are available to A2A through their Bloomberg subscription. Among these, the German electricity market, known as European Energy Exchange (EEX), is the most liquid and provides the highest number of listed options. It is thus chosen as the source of the data.

To make the test more complete and effective both a single-factor mean-reverting model (see Section 3.3.2) and a two-factor LMR-GW model are tested.

Historical calibration trough Kalman filter and maximum likelihood is implemented as described in [23]. The code developed by Alice Guerini and Andrea Marziali in the context of [2] is used. The time horizon of the calibration is set to 3 years from the day when the option prices are taken.

Market calibration, on the other hand, is implemented as described in the previous section. Option pricing is carried out through the Black-Scholes formula 5.1.1, where the variance of the underlying is estimated trough the analytical solution of the Lyapunov equation, both for the single factor and the two-factors models. Initial values for the variance matrix P are set to zero: several trials have shown that they have little influence, as the optimization process tunes the others parameters accordingly.

The risk-free interest rate r required by the Black-Scholes formula is also set to zero. It is common opinion that the best approximations of the risk-free interest rate are either the interest rates of central banks or the premium paid by government bonds of solid countries - for example, Germany. On 1 October 2017, the interest rate of the European Central Bank is -0.4% while the yield of the 1-year Bund is -0.71%. A null risk-free interest rate seems thus adequate.

The numerical optimization is carried out by the `fmincon` Matlab function, which is based on the interior points algorithm. This procedure is chosen because, according to Matlab documentation, it is the most robust in minimizing non-linear and non-quadratic functions.

5.4.1 Test case

Three days have been chosen: 13/02/2017 and 14/02/2017 share the highest number of listed options in the year, 66, while the other, 31/07/2017, has the lowest amount of listed options, 23. From the three datasets, training and validation data are extracted randomly. The random choice has two positive effects. First, it makes the tests on both 13/02 and 14/02 significant: options prices do not vary a lot from day to day, but the random sampling makes the two training and validation datasets completely different. Second, it allows the training dataset to cover evenly all ranges of strikes and maturities.

It is worth noting that the distinction between training and validation datasets is critical only when market calibration is considered. Historical calibration does not take into account the options listed in the same day when pricing is performed: it does not gain any advantage from the training dataset. Market calibration, on the other hand, fits model parameters on the listed options: it is thus important to put it to the test on a different dataset from the training set.

The following table summarizes the most important features of the datasets: validation datasets are slightly different in size because of the adopted sampling procedure.

Day	**Training options**	**Validation options**	**Validation (%)**
13/02/2017	48	18	27%
14/02/2017	55	11	17%
31/07/2017	18	5	22%

Table 5.1: Summary of training and validation datasets.

5.4.2 Calibration

The results of both historical and market calibration are presented in the following tables.

Model	Calibration	μ	λ	σ_1	σ_2
Single factor	Historical	-0.0547	295.9171	10.5868	N/A
Single factor	Market	N/A	296.0266	4.4307	N/A
Two factors	Historical	0.0441	202.2662	7.0498	0.2329
Two factors	Market	N/A	202.3670	1.9380	0.2449

Table 5.2: Calibrated parameters for 13/02/2017.

Model	Calibration	μ	λ	σ_1	σ_2
Single factor	Historical	-0.0542	296.0254	10.5991	N/A
Single factor	Market	N/A	296.1372	4.3226	N/A
Two factors	Historical	0.0424	202.3809	7.0605	0.2327
Two factors	Market	N/A	202.4805	1.9403	0.2451

Table 5.3: Calibrated parameters for 14/02/2017.

Model	Calibration	μ	λ	σ_1	σ_2
Single factor	Historical	-0.0673	302.8399	12.7795	N/A
Single factor	Market	N/A	303.0422	3.0778	N/A
Two factors	Historical	0.0253	199.5133	8.8211	0.1166
Two factors	Market	N/A	206.4031	0.9172	0.1960

Table 5.4: Calibrated parameters for 31/07/2017.

In all the tables, N/A denotes Not Applicable parameters. Specifically, a single parameter for volatility is required by single-factor models and the trend parameter μ cannot be calibrated against option prices because they do not depend on it.

As one may expect, μ does not change a lot from one day to another. It is related to the long-term trend of the prices, so it changes very slowly. Interestingly enough, μ from the single-factor model and from the two-factors model have opposite signs. Bayesian estimation is launched to further investigate the fact: the estimated distribution has significant tails both in the positive and in the negative semiaxes. This happens because German electricity prices between 2014 and 2017 did not show a clear trend, alternating seasons of growth to periods of decrease.

The parameter which accounts for the strength of the mean reversion, λ, takes always very high values, indicating that the spikes from the long-time trend vanish vary fast. This fact means that all the terms multiplied by the exponential $e^{-\lambda t}$ and $e^{-2\lambda t}$ do not give any substantial contribute the the overall variance of the underlying. Given the type of model, market and historical calibrations generally agree on the value of λ.

The σ_1 and σ_2 parameters, which account for the volatility of the underlying are the most different between historical and market calibration. The greatest difference is recorded on σ_1. In case of a single-factor model, σ_1 takes into account all the volatility of the underlying, while, when a two-factor model is concerned, it is linked to short-term volatility. In all the three days, market calibration results in a way smaller short-term volatility then historical calibration.

The previous considerations about volatilities do not imply that one of the approaches is wrong. The opinion of the market, which is expressed by the price of options, is different from the statistical evidence from past data. As the next section shows, this difference has an important impact on the ability of the calibrated model of providing accurate pricing.

5.4.3 Results

The figure of merit for pricing accuracy is the Mean Absolute Error (MAE). It is preferred over Mean Absolute Percentage Error (MAPE) because it is not influenced by the actual price value. As the prices of some options are very close to zero, a relative measure of the error would penalize too heavily poor accuracy on those items.

MAE is defined by the following formula:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|P_i - \hat{P}_i\right| \tag{5.4.1}$$

where n is the total number of priced options, P_i is the actual market price of the i-th option and $\hat{P}_i$ is the price of the same option given by the Black-Scholes formula.

The following tables show MAEs achieved by different models and approaches.

Model	**Calibration**	**Training**	**Validation**
Single factor	Historical	2.5593	2.5955
Single factor	Market	0.4630	0.3487
Two factors	Historical	2.0183	1.9659
Two factors	Market	0.0686	0.0497

Table 5.5: Mean absolute errors for 13/02/2017.

Model	**Calibration**	**Training**	**Validation**
Single factor	Historical	2.5172	2.7768
Single factor	Market	0.4184	0.4905
Two factors	Historical	1.9696	2.1301
Two factors	Market	0.0659	0.0649

Table 5.6: Mean absolute errors for 14/02/2017.

Model	**Calibration**	**Training**	**Validation**
Single factor	Historical	4.5285	3.8627
Single factor	Market	0.2581	0.2887
Two factors	Historical	3.6361	3.0918
Two factors	Market	0.0854	0.1412

Table 5.7: Mean absolute errors for 31/07/2017.

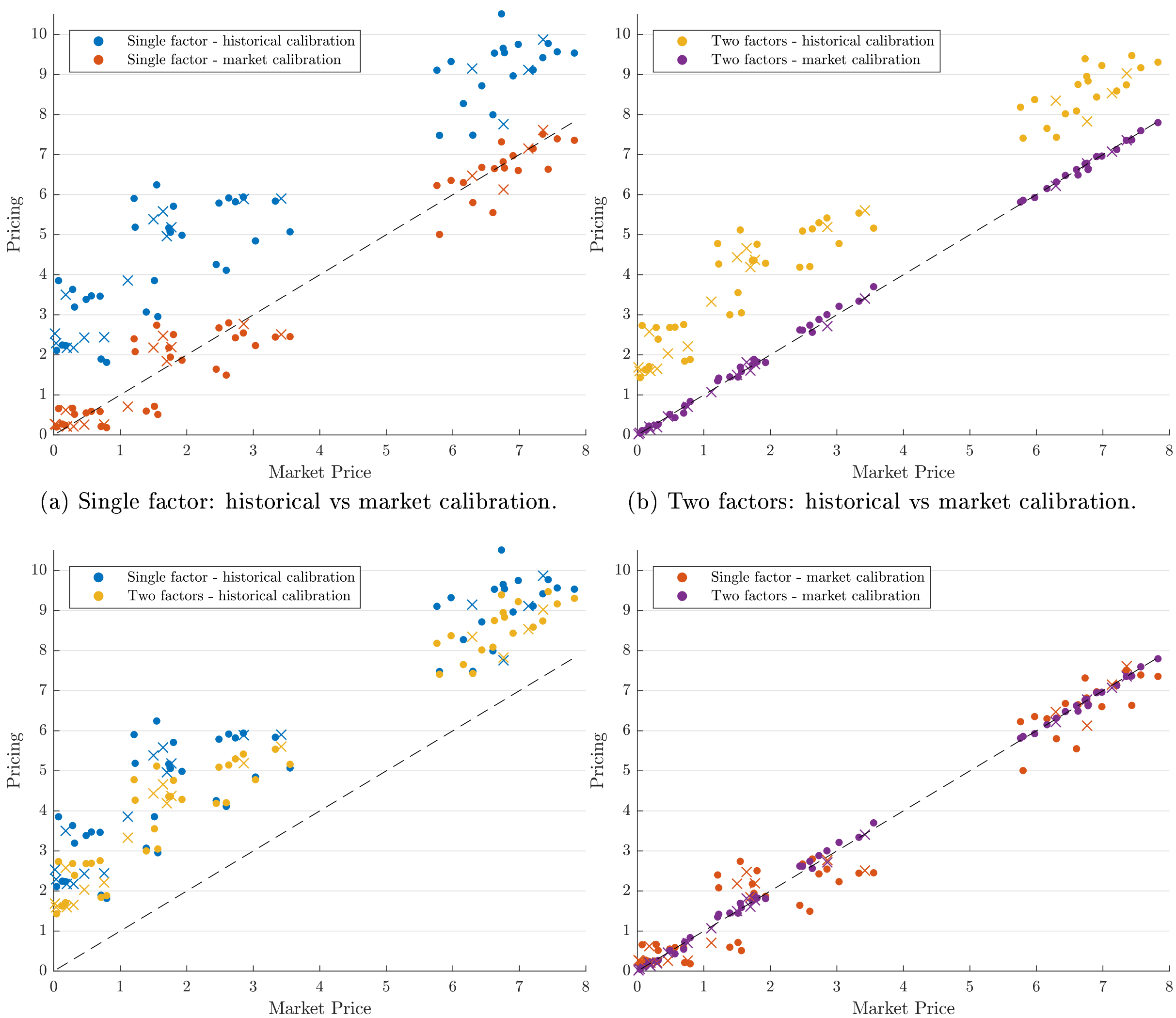
The leitmotiv is quite evident: the two-factor model outperforms the single-factor model; market calibration is more effective than historical calibration.

As expected, the models calibrated on historical data do not suffer any performance deterioration between training and validation datasets, as they are not trained on listed options. Quite interestingly, also the models calibrated against market prices show comparable performances between test and validation datasets, with the only exception of the two-factor model on 31/07. However, this test case is the toughest, with only 18 options in the training set and 5 in the validation set.

To better analyse the performances and to point out the strengths and weaknesses of different approaches, the case of 13 /02 is analysed in detail in the following section.

5.4.4 Remarks

To get a visual picture of how close the prices extracted from different models are to the market, scatter plots are drawn.



(a) Single factor: historical vs market calibration.

(b) Two factors: historical vs market calibration.

(c) Historical calibration: single factor vs two factors.

(d) Market calibration: single factor vs two factors.

Figure 5.4.1: Model-based pricing and market prices on 13/02.

The market price, here considered the golden truth, lies on the horizontal axis, while the inferred prices lie on the vertical axes. To ease the reading of the pictures, a 45-degree dashed line is added to the charts. The equation of that line is $y = x$, that is inferred prices equal to market prices. The more the prices extracted from a model are close to that line, the most the model can follow the market.

In all the charts, circles denotes items from the training set while crosses items from the validation set.

Figure 5.4.1a and 5.4.1b compare the performance of the same model which has been calibrated with either historical or market data. The scatter plot highlights a systematic error of the model trained on historical data. This can be explained with the different values of σ_1. To better understand the effect of the volatility σ on the price of an option - suppose the price of a call option c - it may be worth defining

$$s = \sqrt{p(T)} \tag{5.4.2}$$

and computing the sensitivity of the option price on the standard deviation of the underlying. If the standard Black-Scholes formula based on a geometric Brownian motion model were used, it would be a known function, called Vega. However, as a different model is adopted, the derivation of the sensitivity function has to be carried out.

$$\frac{\partial c}{\partial s} = \frac{\partial}{\partial s}\left\{S_0 N(d_1) - Ke^{-rT} N(d_2)\right\} \tag{5.4.3}$$

where:

$$\begin{aligned} d_1 &= \frac{\ln(S_0/K) + rT + \frac{1}{2}s^2}{s} \\ d_2 &= \frac{\ln(S_0/K) + rT - \frac{1}{2}s^2}{s} = d_1 - s \end{aligned} \tag{5.4.4}$$

One has:

$$\begin{aligned} \frac{\partial c}{\partial s} &= S_0 \frac{\partial N(d_1)}{\partial d_1}\frac{\partial d_1}{\partial s} - Ke^{-rT}\frac{\partial N(d_2)}{\partial d_2}\frac{\partial d_2}{\partial s} \\ &= S_0 N'(d_1)\frac{-d_2}{s} - Ke^{-rT}N'(d_2)\left(\frac{-d_1}{s}\right) \\ &= \frac{1}{\sqrt{2\pi}}\left(S_0 e^{-d_1^2/2}\frac{-d_2}{s} + Ke^{-rT}e^{-d_2^2/2}\frac{d_1}{s}\right) \\ &= \frac{1}{\sqrt{2\pi}}e^{-d_1^2/2}\left(S_0\frac{-d_2}{s} + Ke^{-rT}e^{\left(d_1^2-d_2^2\right)/2}\frac{d_1}{s}\right) \\ &= N'(d_1)\,S_0 \end{aligned} \tag{5.4.5}$$

The result holds also when a put option is considered. As N' denotes the Gaussian probability distribution and the price S_0 cannot be negative, the first derivative of the price with respect to the standard deviation of the underlying is strictly positive:

$$\frac{\partial c}{\partial s} = N'(d_1)\,S_0 > 0 \tag{5.4.6}$$

It means that the price of an option is monotonic increasing in the standard deviation of the underlying: the higher the standard deviation $s = \sqrt{p(T)}$, the higher the price. As in both the one-factor and the two-factor models $p(t)$ is also monotonic increasing in σ_1, the difference in the values of this parameter may be a good explanation for the systematic error of the model trained by historical calibration. Throughout all the three days, historical calibration leads to higher values of σ_1, thus producing higher prices.

Another interesting point is highlighted by Figure 5.4.1d, which compares one-factor and two-factor models both trained with market calibration. Prices produced by the two-factor model show a smaller dispersion from the 45-degree line. This phenomenon is not easy to explain due to the complexity of Black-Scholes formula and the numerical optimization algorithm. A reasonable explanation may reside in the different evolution of the variance of the two models. As shown in Section 3.3, while the variance of a single-factor model flattens after few time units - few days with such an high λ - the variance on a two-factor model does not and follows a linear path.

A time-varying variance may help the optimization algorithm to fit the parameters to options with different maturities, thus keeping the price produced by the two-factor model closer to the market. Moreover, the presence of two different parameters to control short-term and long-term variance surely improves the fitting.

Overall, the benchmark shows that a two-factor model trained with market calibration may be the best choice to get close to the market prices. Pricing non-listed options is a good test, as it is one of the practical task the models should accomplish: however, it is not the only one. Models may also be calibrated to price more complex derivative, relying on several underlyings, as the following section explains.

5.5 Pricing complex derivatives

Plan vanilla options are one of the simplest derivative. Over the last decades, more and more complex derivatives have been developed. They often rely on several underlyings and their payoff functions may be highly non-linear. If this is the case, no closed-form solution exist to achieve their pricing and Monte Carlo simulation is adopted.

Monte Carlo (MC) simulations are numerical methods based on the repeated sampling of the same phenomenon. In the context of derivative pricing, MC is used to simulate thousands of paths for the underlyings, each path being a single realization of the same stochastic process. Every simulated path corresponds to a possible evolution of an underlying. An example of a Monte Carlo simulation of 500 paths of a Geometric Brownian Motion is provided in the Figure 5.5.1.

After a path is generated, the payoff of the derivative is computed on the result of the simulation. Payoffs obtained in all the simulations are then averaged in order to get the most reliable value. If the amount of trials is large enough, the final probability distribution should account for all the possible behaviours of the underlying. Such an approach thus enables to compute the final probability distribution of the payoff, even when its analytical expression is impossible to derive.

MC simulation is a very powerful tool: however, it relies on models. Models for the underlying have to be carefully chosen and, most notably, precisely calibrated. This is why, before entering the complex derivative pricing, the models for the underlying are calibrated against the market, leveraging the procedures described in the previous sections. Usually, markets of vanilla options are exploited.

A real-world example of a complex derivative is the Virtual Power Plant,which is described by the following section.

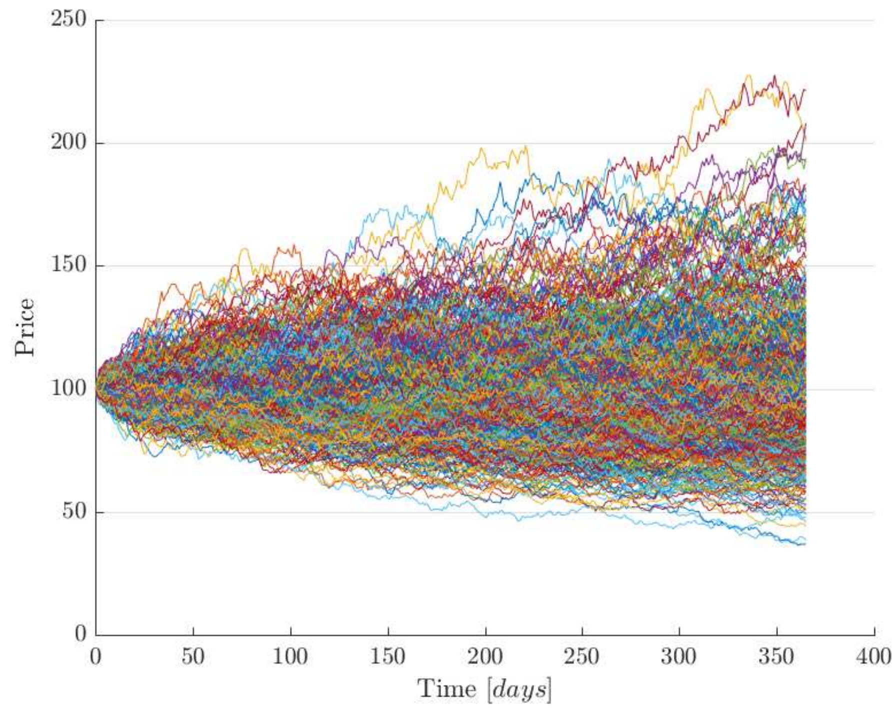


Figure 5.5.1: Monte Carlo simulation of 500 paths of a Geometric Brownian Motion with $\mu = 0$ and $\sigma = 0.015$.

5.5.1 Virtual Power Plant

According to the European Union Electricity Market Glossary [7], a Virtual Power Plant (VPP) is described by the following features.

> Under a virtual power plant scheme the incumbent party is obliged to sell some of his generation capacity to third party market participants (e.g. new entrants). The buyer of the VPP contract has the option to consume power of his VPP against the agreed virtual production cost, but not the obligation and hence the contract can be seen as a call option. VPPs have been imposed by competition authorities in Europe both as a remedy in merger cases and to address dominance.

To model an asset as complex as a power plant, a VPP relies on several underlyings. If a thermal power plant is considered, the following factors impact the value of a VPP:

- All the costs c of the plant: maintenance, management, personnel, etc.
- The price E of the electricity.
- The price F of the fuel.
- The price C of the carbon dioxide emissions.

This last item requires a bit of explanation. In an attempt of boost the adoption of renewable energy source, is 2005 the European Union introduced the Emissions Trading Scheme. All the power facilities has to buy the right of emitting greenhouse gas through ad-hoc financial products, named carbon credits. Carbon credits are listed on markets and can be traded: the more a power plant wants to emit carbon dioxide, the more credits it has to buy.

The value of a VPP from between a certain time $t = 0$ and a maturity $t = T$ is:

$$V = \max_{q \in \Omega} \mathbb{E} \left[\sum_{t=0}^{T} \left(e^{-r_t t} \left(q_t E_t - H\left(q_t\right) F_t - q_t C_t \right) - c\left(q_t\right) \right) \right] \tag{5.5.1}$$

where

- q_t is the operating decision at time t, that is if the plant works or not at each time step. The decision has to be taken among a finite set Ω
- r_t denotes the risk-free interest rate for the time t
- $H\left(q_t\right)$ denotes the calorific power of the gas as a function of the control decision

Finding the value of V requires MC simulation to take care of the expectation operator and dynamic programming algorithms to solve the optimization problem.

Without diving too deep into the detail, it is worth noting that both E, F and C can be modelled by a LMR-GW model. To properly solve the pricing problem, the model should be calibrated against available market prices. Recently, A2A has stated the adoption of the procedure described in this chapter in order to price Virtual Power Plants.

Part III

Beyond vanilla options

Chapter 6

Spread options

Spread options are slightly more complex than vanilla options, as they derive their value from the difference in price between two underlyings. An overview on spread options are their utility in the energy field is provided in Section 1.4.2, so that this chapter can start by defining a mathematical model for the underlyings. As common sense suggests, the underlyings may be correlated and the model should take this fact into account.

Once the construction of the model is completed, the same path of the vanilla options is followed.

The model is translated into a linear stochastic system.

An analytical expression for the time-varying variance of the underlying is derived through the Lyapunov equation.

The result is plugged into the Margrabe's formula to get the pricing of a spread option.

Unfortunately, it is not possible to replicate the approach followed in Chapter 5. Spread options are traded mostly over the counter and there is no market to directly calibrate the model against. However, all model parameters but the correlation between the underlyings can be calibrated against vanilla option markets.

In developing the computations required to get to the variance of the two-underlying model, some lemmas are introduced to generalize the steps followed for a single-underlying model. There is no reason to limit these results to two underlyings, as they can be easily generalized to n underlyings, thus paving the way to applications on the Value at Risk (VaR) of an arbitrarily large portfolio.

6.1 A model for two underlyings

A mathematical model for the case of two correlated underlyings may be developed by extending the set of equations of a LMR-GW model 3.4.1. Both the log-spot prices x and y follow a two factor model, with independent Wiener processes driving the short-term and the long-term components.

However, the long-term trends of the two underlying and the short-term diversions of the log-spot prices are correlated. To better understand why this happens, two real-world scenarios are presented.

Suppose that x is the spot price of electricity in Italy and y the same asset in German. If the long-term trend of x is increasing, say, because of an increase in the price of fuels, it is clear that the trend of y cannot be very different. Pretend now that the price of x has a short-term strong increase. For electricity providers in Germany is then more profitable to sell their product in Italy: they divert part of the production to Italy, thus lowering home offer and increasing German price y.

On the other hand, there is no evidence of a strong correlation between the long-term trend of one asset and the short-term behaviour of the other. This is compliant with the assumption of the LMR-GW model, where long-term and short-term components are independent, even where the same asset is considered.

The whole model is described by the following set of stochastic differential equations:

$$\begin{cases} s_x(t) = e^{x_1(t)} \\ s_y(t) = e^{y_1(t)} \\ dx_1(t) = \lambda_x (x_2(t) - x_1(t))\, dt + \sigma_{x1} dW_{x1}(t) \\ dx_2(t) = \mu_x dt + \sigma_{x2} dW_{x2}(t) \\ dy_1(t) = \lambda_y (y_2(t) - y_1(t))\, dt + \sigma_{y1} dW_{y1}(t) \\ dy_2(t) = \mu_y dt + \sigma_{y2} dW_{y2}(t) \end{cases} \tag{6.1.1}$$

where all the variables and parameters have the same meaning that the ones in the single-underlying LMR-GW model.

6.2 Transformation to a stochastic linear model

As in the single-underlying case, the model can be written in the form of a stochastic linear system

$$\begin{cases} \frac{dx(t)}{dt} = Ax(t) + Bu(t) \\ y(t) = Cx(t) \end{cases} \tag{6.2.1}$$

by performing the following operations:

1. Divide all the equations by the time differential dt
2. Define the state vector x and the input vector u

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ y_1(t) \\ y_2(t) \end{bmatrix}, \quad u(t) = \begin{bmatrix} \frac{dW_{x1}(t)}{dt} \\ \frac{dW_{x2}(t)}{dt} \\ \frac{dW_{y1}(t)}{dt} \\ \frac{dW_{y2}(t)}{dt} \end{bmatrix} \sim WN(0, S) \tag{6.2.2}$$

where S is the symmetric, positive definite matrix defined as follows:

$$S = \begin{bmatrix} 1 & 0 & \rho_s & 0 \\ 0 & 1 & 0 & \rho_l \\ \rho_s & 0 & 1 & 0 \\ 0 & \rho_l & 0 & 1 \end{bmatrix} \tag{6.2.3}$$

According to the properties of white noise, choosing S is equivalent to fixing the covariance matrix of the process. It holds:

$$Cov[u(t_1), u(t_2)] = S\delta(t_2 - t_1), \ \forall t_1, t_2 \tag{6.2.4}$$

The covariance matrix has non-zero elements only on the main diagonal and in the entries which account for the correlation between long-term and short-term components of the two underlyings.

3. Set the matrices A, B, and C accordingly:

$$A = \begin{bmatrix} -\lambda_x & \lambda_x & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -\lambda_y & \lambda_y \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{6.2.5}$$

$$B = \begin{bmatrix} \sigma_{x1} & 0 & 0 & 0 \\ 0 & \sigma_{x2} & 0 & 0 \\ 0 & 0 & \sigma_{y1} & 0 \\ 0 & 0 & 0 & \sigma_{y2} \end{bmatrix} \tag{6.2.6}$$

$$C = \begin{bmatrix} 1 & 0 & -1 & 0 \end{bmatrix} \tag{6.2.7}$$

The C matrix is defined so that the output of the linear system is

$$y(t) = x_1(t) - y_1(t) \tag{6.2.8}$$

that is the difference in price between the two underlyings.

6.3 Variance estimation: solution to the Lyapunov equation

Starting from the spread options model 6.1.1 written as a liner stochastic system, it is possible to compute the variance of the state by using once more the solution to the matrix differential Lyapunov equation 4.1.3. One has thus to compute:

$$P(t) = e^{At} P_0 e^{A^T t} + \int_0^t e^{A(t-z)} B S B^T e^{A^T (t-z)} dz \tag{6.3.1}$$

The same steps which have been presented in Section 4 will be followed.

6.3.1 Matrix exponential

The first problem one has to face is the computation of the matrix exponential e^{At}. As the matrix A now belongs to $M_{\mathbb{R}}^{4\times 4}$, the approach previously presented would be more demanding.

Even worse, A has an eigenvalue with molteplicity 2. So, the eigenvalues criterion cannot guarantee that A is diagonalizable. Fortunately, the following lemma solves this problem.

Lemma 6 (Diagonalizaility of block-diagonal matrices).
If matrix M is block-diagonal and all the blocks are diagonalizable, then M is diagonalizable.

Proof. The proof will be developed in the case when the block-diagonal matrix M is made of two diagonalizable matrices M_1 and M_2. The result can be easily generalized.

As A_1 and A_2 are diagonalizable, there exist two invertible matrices U_1, U_2 and two diagonal matrices D_1 and D_2 such that:

$$\begin{aligned} D_1 &= U_1^{-1} M_1 U_1 \\ D_2 &= U_2^{-1} M_2 U_2 \end{aligned} \tag{6.3.2}$$

It is now possible to define

$$U := \begin{bmatrix} U_1 & 0 \\ 0 & U_2 \end{bmatrix} \tag{6.3.3}$$

which is invertible, as it holds:

$$U^{-1} = \begin{bmatrix} U_1^{-1} & 0 \\ 0 & U_2^{-1} \end{bmatrix} \tag{6.3.4}$$

Now, the matrix

$$D := U^{-1}MU = \begin{bmatrix} U_1^{-1}M_1U_1 & 0 \\ 0 & U_2^{-1}M_2U_2 \end{bmatrix} = \begin{bmatrix} D_1 & 0 \\ 0 & D_2 \end{bmatrix} \tag{6.3.5}$$

is diagonal, as it is block-diagonal and the blocks are diagonal matrices. □

Matrix A can be written as a block-diagonal matrix:

$$A = \begin{bmatrix} -\lambda_x & \lambda_x & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -\lambda_y & \lambda_y \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} -\lambda_x & \lambda_x \\ 0 & 0 \end{bmatrix} & 0 \\ 0 & \begin{bmatrix} -\lambda_y & \lambda_y \\ 0 & 0 \end{bmatrix} \end{bmatrix} := \begin{bmatrix} A_x & 0 \\ 0 & A_y \end{bmatrix} \tag{6.3.6}$$

and matrices A_1 and A_2 have been proven to be diagonalizable in Section 4.

Now, the problem of computing the matrix exponential becomes a lot easier. One can take advantage of another linear algebra lemma.

Lemma 7 (Exponential of a block-diagonal matrix).
Let M be a block-diagonal matrix:

$$M := \begin{bmatrix} M_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & M_n \end{bmatrix} \tag{6.3.7}$$

Then:

$$e^M = \begin{bmatrix} e^{M_1} & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & e^{M_n} \end{bmatrix} \tag{6.3.8}$$

Proof. The proof straightforwardly follows from the very definition of matrix exponential as a power series. In fact, the powers of a diagonal matrix are diagonal as well. □

One can now exploit the computation performed in Section 4 to get to the expression of e^{At}:

$$e^{At} = \begin{bmatrix} e^{A_xt} & 0 \\ 0 & e^{A_yt} \end{bmatrix} = \begin{bmatrix} e^{-\lambda_x t} & 1 - e^{-\lambda_x t} & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & e^{-\lambda_y t} & 1 - e^{-\lambda_y t} \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6.3.9}$$

6.3.2 First item of the formula

Let's assume P_0 is a diagonal matrix:

$$P_0 = \begin{bmatrix} \begin{bmatrix} p_{x1} & 0 \\ 0 & p_{x2} \end{bmatrix} & 0 \\ 0 & \begin{bmatrix} p_{y1} & 0 \\ 0 & p_{y2} \end{bmatrix} \end{bmatrix} = \begin{bmatrix} P_{0x} & 0 \\ 0 & P_{0y} \end{bmatrix} \tag{6.3.10}$$

By writing both e^{At} and P_0 as block-diagonal matrices, it is possible to simplify the computation of the first item of the Lyapunov equation solution

$$e^{At} P_0 e^{A^T t} = \begin{bmatrix} e^{A_x t} & 0 \\ 0 & e^{A_y t} \end{bmatrix} \begin{bmatrix} P_{0x} & 0 \\ 0 & P_{0y} \end{bmatrix} \begin{bmatrix} e^{A_x^T t} & 0 \\ 0 & e^{A_y^T t} \end{bmatrix} = \begin{bmatrix} e^{A_x t} P_0 e^{A_x^T t} & 0 \\ 0 & e^{A_y t} P_0 e^{A_y^T t} \end{bmatrix} \tag{6.3.11}$$

where, following the computations of Chapter 4:

$$e^{A_x t} P_0 e^{A_x^T t} = \begin{bmatrix} (p_{x1} + p_{x2}) e^{-2\lambda_x t} - 2p_{x2} e^{-\lambda_x t} + p_{x2} & p_{x2} \left(1 - e^{-\lambda_x t}\right) \\ p_{x2} \left(1 - e^{-\lambda_x t}\right) & p_{x2} \end{bmatrix} \tag{6.3.12}$$

$$e^{A_y t} P_0 e^{A_y^T t} = \begin{bmatrix} (p_{y1} + p_{y2}) e^{-2\lambda_y t} - 2p_{y2} e^{-\lambda_y t} + p_{y2} & p_{y2} \left(1 - e^{-\lambda_y t}\right) \\ p_{y2} \left(1 - e^{-\lambda_y t}\right) & p_{y2} \end{bmatrix} \tag{6.3.13}$$

As expected, the solution is made of two blocks, each of them exactly identical to the result 4.3.6, apart from the different definition of matrix P_0.

6.3.3 Second item of the formula

As the S matrix is full, there is no easy way of performing the matrix products which lead to the expression of the second item. The only simplification comes from a change of integration variable, resembling what has already been presented in Chapter 4.

$$\int_0^t e^{A(t-z)} BSB^T e^{A^T (t-z)} dt = \int_0^\tau e^{A\tau} BSB^T e^{A^T \tau} d\tau \tag{6.3.14}$$

To avoid human errors, brute force computations are carried on with the help of a symbolic tool. The result is the following symmetric matrix:

$$\int_0^\tau e^{A\tau} BSB^T e^{A^T \tau} d\tau = \begin{bmatrix} R_{11} & R_{12} & R_{13} & R_{14} \\ & R_{22} & R_{23} & R_{24} \\ & & R_{33} & R_{34} \\ & & & R_{44} \end{bmatrix} \tag{6.3.15}$$

where:

$$R_{11} = -\frac{\sigma_{x2}^2 + \sigma_{x1}^2}{2\lambda_x} e^{-2\lambda_x t} + \frac{2\sigma_{x2}^2}{\lambda_x} e^{-\lambda_x t} + \sigma_{x2}^2 t + \frac{\sigma_{x1}^2 - 3\sigma_{x2}^2}{2\lambda_x}$$

$$R_{22} = \sigma_{x2}^2 t$$

$$R_{33} = -\frac{\sigma_{y2}^2 + \sigma_{y1}^2}{2\lambda_y} e^{-2\lambda_y t} + \frac{2\sigma_{y2}^2}{\lambda_y} e^{-\lambda_y t} + \sigma_{y2}^2 t + \frac{\sigma_{y1}^2 - 3\sigma_{y2}^2}{2\lambda_y}$$

$$R_{44} = \sigma_{y2}^2 t$$

$$R_{12} = \sigma_{x2}^2 \left(t + \frac{1 - e^{-\lambda_x t}}{\lambda_x}\right)$$

$$R_{13} = \rho_2 \sigma_{x2} \sigma_{y2} \left(t + \frac{e^{-\lambda_x t} - 1}{\lambda_x} + \frac{e^{-\lambda_y t} - 1}{\lambda_y}\right) - (\rho_1 \sigma_{x1} \sigma_{y1} + \rho_1 \sigma_{x2} \sigma_{y2}) \frac{e^{-(\lambda_x + \lambda_y)t} - 1}{\lambda_x + \lambda_y}$$

$$R_{14} = \frac{\rho_2 \sigma_{x2} \sigma_{y2} \left(e^{-\lambda_x t} + \lambda_x t - 1\right)}{\lambda_x}$$

$$R_{23} = \frac{\rho_1 \sigma_{x2} \sigma_{y2} \left(e^{-\lambda_y t} + \lambda_y t - 1\right)}{\lambda_y}$$

$$R_{24} = \rho_2 \sigma_{x2} \sigma_{y2} t$$

$$R_{34} = \sigma_{y2}^2 \left(t + \frac{1 - e^{-\lambda_y t}}{\lambda_y}\right)$$

6.3.4 Final result

It is now possible to add up the two items of the Lyapunov equation solution to get to the final formulation of $P(t)$, which is the following symmetric matrix:

$$P(t) = \begin{bmatrix} P_{11} & P_{12} & P_{13} & P_{14} \\ & P_{22} & P_{23} & P_{24} \\ & & P_{33} & P_{34} \\ & & & P_{44} \end{bmatrix} \tag{6.3.16}$$

where:

$$P_{11} = \left(p_{x2} + p_{x1} - \frac{\sigma_{x2}^2 + \sigma_{x1}^2}{2\lambda_x}\right) e^{-2\lambda_x t} + \left(\frac{2\sigma_{x2}^2}{\lambda_x} - 2p_{x2}\right) e^{-\lambda_x t} + \sigma_{x2}^2 t + \frac{\sigma_{x1}^2 - 3\sigma_{x2}^2}{2\lambda_x} + p_{x2}$$

$$P_{22} = \sigma_{x2}^2 t + p_{x2}$$

$$P_{33} = \left(p_{y2} + p_{y1} - \frac{\sigma_{y2}^2 + \sigma_{y1}^2}{2\lambda_y}\right) e^{-2\lambda_y t} + \left(\frac{2\sigma_{y2}^2}{\lambda_y} - 2p_{y2}\right) e^{-\lambda_y t} + \sigma_{y2}^2 t + \frac{\sigma_{y1}^2 - 3\sigma_{y2}^2}{2\lambda_y} + p_{y2}$$

$$P_{44} = \sigma_{y2}^2 t + p_{y2}$$

$$P_{12} = \left(\frac{\sigma_{x2}^2}{\lambda_x} - p_{x2}\right) e^{-\lambda_x t} + \sigma_{x2}^2 t + p_{x2} - \frac{\sigma_{x2}^2}{\lambda_x}$$

$$P_{13} = \rho_2 \sigma_{x2} \sigma_{y2} \left(t + \frac{e^{-\lambda_x t} - 1}{\lambda_x} + \frac{e^{-\lambda_y t} - 1}{\lambda_y}\right) - (\rho_1 \sigma_{x1} \sigma_{y1} + \rho_1 \sigma_{x2} \sigma_{y2}) \frac{e^{-(\lambda_x + \lambda_y)t} - 1}{\lambda_x + \lambda_y}$$

$$P_{14} = \frac{\rho_2 \sigma_{x2} \sigma_{y2} \left(e^{-\lambda_x t} + \lambda_x t - 1\right)}{\lambda_x}$$

$$P_{23} = \frac{\rho_1 \sigma_{x2} \sigma_{y2} \left(e^{-\lambda_y t} + \lambda_y t - 1\right)}{\lambda_y}$$

$$P_{24} = \rho_2 \sigma_{x2} \sigma_{y2} t$$

$$P_{34} = \left(\frac{\sigma_{y2}^2}{\lambda_y} - p_{y2}\right) e^{-\lambda_y t} + \sigma_{y2}^2 t + p_{y2} - \frac{\sigma_{y2}^2}{\lambda_y}$$

As one may expect, items on the main diagonal resembles the result found for a single-underlying model.

While the variance of a single-underlying model was analysed - Section 4.4, it was found that all the terms were made of two components. A long-term variance asymptotic to t and a short-term contribution, vanishing fast under the effect of the exponential terms. Both P_{11} and P_{33} show the same structure. Interestingly enough, the covariance terms are also composed in the same way. The only remarkable difference resides in the coefficients of the long-term components, which are proportional to the correlation parameters ρ_1 and ρ_2.

6.4 Analytical solution and Monte Carlo simulation

As shown in the previous chapter, Monte Carlo simulation is a useful tool to empirically measure values for which an analytical expression is either impossible or very hard to derive. For example, if no solution could be found for the Lyapunov equation, a MC simulation would be the only way to get to the time-varying covariance matrix of the two underlyings.

It may thus be interesting to compare the results found by the MC simulations with the analytical solution. A complete match, except for the unavoidable error carried by the random component of MC methods, would be a proof of the correctness for the analytical solution. In this respect, a match between the analytical and the simulated results is equivalent to the substitution of the result in the original equation performed in Chapter 4.

As the test is on the model itself and does not include model-based pricing, parameters do not have to be realistic. Quite arbitrarily, the values reported in Table 6.1 are chosen.

Parameter	Value
λ_x	0.5
λ_y	0.4
ρ_1	0.3
ρ_2	0.6
σ_{x1}	0.5
σ_{x2}	0.15
σ_{y1}	0.35
σ_{y2}	0.1
μ_x	0.01
μ_y	-0.01

Table 6.1: Parameters of the two-underlying model.

Daily log-spot prices are simulated for 365 days, with an initial values for both the underlyings of 100. In the Monte Carlo simulation, a total of 10000 simulations were performed. Figure 6.4.1 represents the realizations of the first underlying x. It is evident effect of the long-term component of the standard deviation, which makes the paths spread in a shape which resembles $\sqrt{t}$.

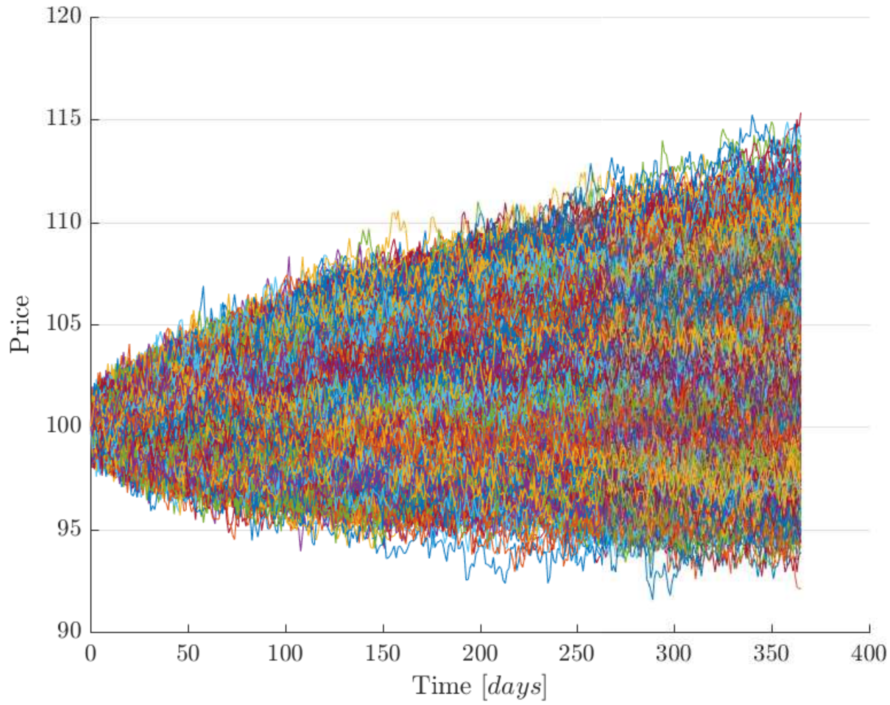


Figure 6.4.1: Monte Carlo simulation of 10000 paths for asset x.

After the simulations have been performed, the variance of all the paths at time final time step $P_{MC}(T)$ can be computed and compared with the analytical solution $P(t)$ evaluated at the same time instant. It is found:

$$P = \begin{bmatrix} 8.3950 & 8.1675 & 3.3128 & 3.2670 \\ 8.1675 & 8.2125 & 3.2625 & 3.2850 \\ 3.3128 & 3.2625 & 3.7656 & 3.6250 \\ 3.2670 & 3.2850 & 3.6250 & 3.6500 \end{bmatrix} \tag{6.4.1}$$

$$P_{MC} = \begin{bmatrix} 8.5050 & 8.1985 & 3.3191 & 3.2505 \\ 8.1985 & 8.2599 & 3.2679 & 3.2854 \\ 3.3191 & 3.2679 & 3.8115 & 3.6302 \\ 3.2505 & 3.2854 & 3.6302 & 3.6568 \end{bmatrix} \tag{6.4.2}$$

Apart from errors introduced by the random nature of MC methods, the result is very similar. To better appreciate the equivalence of the two methods, some elements of the matrices P and P_{MC} are plotted against time. To have on the vertical axis the same dimension of the underlying spot price, the square root is extracted, thus passing from variances to standard deviations.

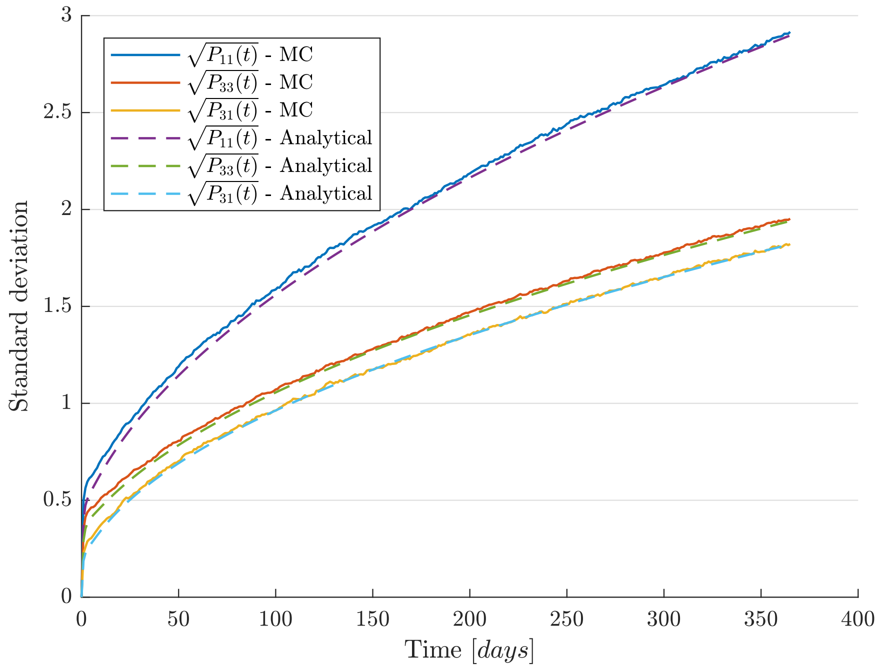


Figure 6.4.2: Evolution of some elements of $\sqrt{P(t)}$ and $\sqrt{P_{MC}(t)}$ against time.

The greatest advantage the analytical solution carries is its efficiency. While computing the variance of the underlying at a single time instant T results in evaluating a function once, performing MC simulations requires thousands of evaluation of the stochastic differential equations of the model. Even worse, the bigger is T, the more evaluations are required.

After the correctness and the efficiency of the analytical solution have been proven, it is possible to dive into its practical application to option pricing.

6.5 Pricing spread options

As shown in Section 2.5.2, if the Black-Scholes assumptions hold, the price of a spread option which allows to exchange a unit of an asset S_1 for a unit of another asset S_2 is:

$$V = S_1(0) N(d_1) - S_2(0) N(d_2) \tag{6.5.1}$$

where:

$$\begin{aligned} d_1 &= \frac{\ln(S_1(0)/S_2(0)) + (\sigma^2/2)T}{\sigma\sqrt{T}} \\ d_2 &= d_1 - \sigma\sqrt{T} \\ \sigma &= \sqrt{\sigma_1^2 + \sigma_2^2 - 2\sigma_1\sigma_2\rho} \end{aligned} \tag{6.5.2}$$

Exactly as in the Black-Scholes formula, all the parameters but σ are established by the contract. Moreover, the same remarks on the term $\sigma\sqrt{T}$ are valid. The item represents the standard deviation of $S_1 - S_2$ at time T when both S_1 and S_2 are modeled by a geometric Brownian motion. If a different model for the underlyings is adopted, d_1 and d_2 in the previous formula should be modified accordingly.

Let $p(t)$ be a time-varying function representing the variance of $S_1 - S_2$ at each time $t > 0$.

$$p(t) := Var\left[S_1(t) - S_2(t)\right] \tag{6.5.3}$$

Then:

$$d_1 = \frac{\ln\left(S_1(0)/S_2(0)\right) + \frac{1}{2}p(T)}{\sqrt{p(t)}}$$
$$d_2 = d_1 - \sqrt{p(t)}$$

If both the underlyings are described by a two-factor LMR-GW model - Equation 6.1.1, the function $p(t)$ can be easily derived from the variance of the states $P(t)$ reported in Equation 6.3.16. By applying the output transformation of the linear system, one has:

$$\begin{aligned} Var\left[S_1(t) - S_2(t)\right] &= Var\left[y(t)\right] = Var\left[Cx(t)\right] \\ &= CVar\left[x(t)\right]C^T = CP(t)C^T \qquad (6.5.4) \\ &= P_{11}(t) - P_{13}(t) - P_{31}(t) + P_{33}(t) \qquad (6.5.5) \end{aligned}$$

Leveraging the symmetry of $P(t)$, one has:

$$p(t) = Var\left[S_1(t) - S_2(t)\right] = P_{11}(t) + P_{33}(t) - 2P_{13}(t) \tag{6.5.6}$$

This formula resembles a well-known result on the variance of the difference of random variables. The elements P_{11} and P_{33} of the P matrix are, respectively, the variance of the state x_1 and the variance of the state y_1, that are the underlyings S_1 and S_2. P_{13}, on the other hand, is the covariance between the two underlyings. So, the function $p(t)$ can also be read as:

$$p(t) = Var\left[S_1(t) - S_2(t)\right] = Var\left[S_1(t)\right] + Var\left[S_2(t)\right] - 2Cov\left[S_1(t), S_2(t)\right] \tag{6.5.7}$$

6.5.1 Model calibration

Two-underlying models which are used to price spread options cannot be directly calibrated on the market. In fact, a real market for spread options does not exist in the energy field. Usually, this kind of derivative is traded over the counter.

However, a two-underlying model is nothing more than two LMR-GW model whose Wiener process are correlated. Thus, all the parameters of the LMR-GW models can be calibrated against the market of vanilla options, as described in Chapter 5.

Only the correlation parameters ρ_1 and ρ_2 cannot be tuned with this procedure. They are usually inferred by computing the correlation of the time series of the two underlyings. To get the short-term correlation, hourly or daily prices are usually considered, while long-term correlation may be inferred from one-year-ahead futures.

6.6 A final summary: how to price a spread option

All the basic tools required to price a spread option have been presented. It is now time to summarize them in a simple workflow.

Assume it is required to price a spread option with payoff $\max\left(S_1\left(T\right)-S_2\left(T\right)\right)$ under the Black-Scholes hypothesis and assume both the underlyings to be well described by a two-factor LMR-GW model. The following steps are followed.

1. Calibrate the two LMR-GW models against vanilla options.
2. Infer the short-term correlation from the time series of the underlyings and the long-term correlation from the prices of futures.
3. Put all the parameters in the analytical solution of the Lyapunov equation to find the variance matrix at maturity.
4. Extract the elements of interest from the variance matrix to compute the variance $p\left(T\right)$ of $S_1\left(T\right)-S_2\left(T\right)$.
5. Plug $p\left(T\right)$ into the Margrabe's formula and get the price of the option.

The derivation of the variance for a two-underlying model is not only important for spread options. With a bit of effort and under some restrictive hypothesis, the results can be generalized to a model with an arbitrary number of underlyings. It is time to face the problem of computing the value at risk of a general portfolio.

Chapter 7

More underlyings: portfolio risk

In the previous chapter the case of a derivative based on two underlying was analysed. There is no reason to limit the number of underlyings: under some restrictive hypothesis, a general approach to the solution of the Lyapunov equation can be developed. Such a procedure may be applied to a wide range of linear stochastic systems.

The first part of the chapter is thus devoted to the derivation and the description of the procedure.

Once the algorithm has been obtained, it is tested against the simple case of a single-underlying LMR-GW model. Results match those derived in Chapter 4.

Then a practical application of the framework is presented. The Value at Risk (VaR) is the maximum loss a portfolio made of several underlyings may suffer with a probability p. After a brief description of the idea of VaR and the methods which enables its computation, the utility of the solution of the Lyapunov equation is shown.

After the pricing of vanilla options and spread options and the calibration of the models required to price complex derivatives, the Value at Risk is the last example of the possible applications of the Lyapunov equation to finance. The next and last chapter summarizes the results of this work and describes some of the possible future developments.

7.1 A general solution to the Lyapunov equation

Let's consider a standard linear stochastic system of arbitrary order. It is described by the following equations, where the definitions and the assumptions stated in Section 3.2 hold:

$$\begin{cases} \frac{dx(t)}{dt} = Ax\left(t\right) + Bu\left(t\right) \\ y\left(t\right) = Cx\left(t\right) \end{cases} \tag{7.1.1}$$

subject to the initials conditions:

$$E\left[x\left(0\right)\right] = x_0 \tag{7.1.2}$$

$$Var\left[x\left(0\right)\right] = P_0 \tag{7.1.3}$$

The variance $P\left(t\right)$ of the state x is given by:

$$P\left(t\right) = e^{At}P_0e^{A^Tt} + \int_0^t e^{A(t-z)}BSB^Te^{A^T(t-z)}dz \tag{7.1.4}$$

According to [13], the only available result in the literature for the general solution of the Lyapunov equation is [28]. More recent papers, including [25], focus only on the discrete-time solution of the Lyapunov equation, which finds application in the commonly used Kalman filter.

However, the result presented by Rome in [28] assumes the A matrix to be asymptotically stable, and relies on this assumption to exclude the possibility of null eigenvalues.

A general solution to the Lyapunov equation which allows for multiple zero eigenvalues is derived in the following section, provided that A is diagonalizable.

7.2 A more general approach

The derivation of the time-varying variance matrix for a two-state system is quite easy (see Chapter 4). The same procedure becomes harder when the number of states doubles (see Chapter 6). It is not difficult to imagine that analytical computation becomes prohibitive when the number of states is unbounded.

A different approach to the solution of the time-varying Lyapunov equation is now introduced to allow the number of states to grow without increasing the complexity of the solution procedure. The price to pay is just giving up the hope to have a single formula to get the covariance matrix of the assets in favour of a more articulated algorithm.

The starting point of the procedure is an arbitrary stochastic linear system, as in Equation 7.1.1.

Instead of deriving directly the time-dependent function which describes the variance of the state, a preliminary transformation of the system is performed.

This step requires the A matrix to be diagonalizable, which is not always the case. However, as far as the stochastic linear system is derived from LMR-GW models, Lemma 6 and the subsequent considerations ensure that this requirement is satisfied.

If matrix A is diagonalizable, there exist an invertible matrix U and a diagonal matrix $\tilde{A}$ such that

$$\tilde{A} = U^{-1}AU \tag{7.2.1}$$

Let's define now a new state vector $\tilde{x}$ as follows:

$$\tilde{x}(t) = U^{-1}x(t) \tag{7.2.2}$$

As U is invertible, one can invert the previous relationship to get:

$$x(t) = U\tilde{x}(t) \tag{7.2.3}$$

By plugging the previous relationship in the system 7.1.1, one can get the following equivalent system:

$$\begin{gathered}\begin{cases}\frac{dU\tilde{x}(t)}{dt} = AU\tilde{x}(t) + Bu(t)\\ y(t) = CU\tilde{x}(t)\end{cases}\\ \begin{cases}\frac{d\tilde{x}(t)}{dt} = U^{-1}AU\tilde{x}(t) + U^{-1}Bu(t)\\ y(t) = CU\tilde{x}(t)\end{cases}\end{gathered} \tag{7.2.4}$$

And by defining the following auxiliary matrices:

$$\tilde{B} = U^{-1}B \tag{7.2.5}$$

$$\tilde{C} = CU \tag{7.2.6}$$

one can return to the general form of a stochastic linear system, which now has a diagonal dynamic matrix $\tilde{A}$:

$$\begin{cases} \frac{d\tilde{x}(t)}{dt} = \tilde{A}\tilde{x}(t) + \tilde{B}u(t) \\ y(t) = \tilde{C}\tilde{x}(t) \end{cases} \tag{7.2.7}$$

subject to the initial conditions:

$$E[\tilde{x}(0)] = \tilde{x}_0 = U^{-1}x_0 \tag{7.2.8}$$

$$Var[\tilde{x}(0)] = \tilde{P}_0 \tag{7.2.9}$$

The relationship between $\tilde{P}_0$ and P_0 will be derived afterwards.

It is now time to plug the matrices of system 7.2.7 into the solution of the matrix differential Lyapunov equation to retrieve the variance $\tilde{P}(t)$ of the state $\tilde{x}(t)$.

$$\tilde{P}(t) = e^{\tilde{A}t}\tilde{P}_0 e^{\tilde{A}^T t} + \int_0^t e^{\tilde{A}(t-z)}\tilde{B}S\tilde{B}^T e^{\tilde{A}^T(t-z)}dz \tag{7.2.10}$$

The usual change of integration variable makes the second term of the formula a bit easier.

$$\tau = t - z \tag{7.2.11}$$

$$\tilde{P}(t) = e^{\tilde{A}t}\tilde{P}_0 e^{\tilde{A}^T t} + \int_0^t e^{\tilde{A}\tau}\tilde{B}S\tilde{B}^T e^{\tilde{A}^T\tau}d\tau \tag{7.2.12}$$

Despite the exponential matrices are diagonal, the fact that S may be full prevents an easy computation of the previous formula.

However, the following remark upon the product of a diagonal matrix by a full matrix holds.

Remark 1 (Product of full and diagonal matrices)**.**
Let M be a full square matrix and D be a diagonal matrix. Let also $m_{\cdot j}$ denote the j-th column of M and $\{M\}_{ij}$ denote the entry (i,j) of matrix M. Then the product DMD^T results in the following formula:

$$\begin{aligned} DMD^T = DMD &= \begin{bmatrix} d_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & d_n \end{bmatrix} \begin{bmatrix} m_{11} & \cdots & m_{1n} \\ \vdots & \ddots & \vdots \\ m_{1n} & \cdots & m_{nn} \end{bmatrix} \begin{bmatrix} d_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & d_n \end{bmatrix} \\ &= \begin{bmatrix} d_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & d_n \end{bmatrix} \begin{bmatrix} d_1 m_{\cdot 1} & d_2 m_{\cdot 2} & \cdots & d_n m_{\cdot n} \end{bmatrix} \\ &= \begin{bmatrix} d_1 d_1 m_{11} & d_1 d_2 m_{11} & \cdots & d_1 d_n m_{1n} \\ d_2 d_1 m_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ d_n d_1 m_{n1} & \cdots & \cdots & d_n d_n m_{nn} \end{bmatrix} \end{aligned} \tag{7.2.13}$$

Thus,

$$\{DMD^T\}_{ij} = D_{ii}D_{jj}M_{ij} = d_i d_j m_{ij} \tag{7.2.14}$$

It is now possible to take advantage from the previous Remark to simplify a lot the formula for a single element of $\tilde{P}(t)$. Let $\tilde{a}_i$ denote the i-th element on the main diagonal of $\tilde{A}$.

$$\begin{aligned}
\tilde{P}_{ij}(t) &= \left\{ e^{\tilde{A}t}\tilde{P}_0 e^{\tilde{A}^T t} + \int_0^t e^{\tilde{A}\tau}\tilde{B}S\tilde{B}^T e^{\tilde{A}^T\tau} d\tau \right\}_{ij} \\
&= \left\{ e^{\tilde{A}t} \right\}_{ii} \left\{ \tilde{P}_0 \right\}_{ij} \left\{ e^{\tilde{A}^T t} \right\}_{jj} + \int_0^t \left\{ e^{\tilde{A}\tau} \right\}_{ii} \left\{ \tilde{B}S\tilde{B}^T \right\}_{ij} \left\{ e^{\tilde{A}^T\tau} \right\}_{jj} d\tau \\
&= \left\{ \tilde{P}_0 \right\}_{ij} e^{(\tilde{a}_i+\tilde{a}_j)t} + \left\{ \tilde{B}S\tilde{B}^T \right\}_{ij} \int_0^t e^{(\tilde{a}_i+\tilde{a}_j)\tau} d\tau
\end{aligned} \tag{7.2.15}$$

The integral in the previous formula cannot be uniquely solved because of the possibility of multiple null eigenvalues in the matrix A - thus multiple null elements in the matrix $\tilde{A}$. Nonetheless, one can write a case-defined function:

$$\int_0^t e^{(\tilde{a}_i+\tilde{a}_j)\tau} d\tau = \begin{cases} \frac{1}{\tilde{a}_i+\tilde{a}_j}\left(e^{(\tilde{a}_i+\tilde{a}_j)t} - 1\right) & \tilde{a}_i \neq 0 \vee \tilde{a}_j \neq 0 \\ t & \tilde{a}_i = \tilde{a}_j = 0 \end{cases} \tag{7.2.16}$$

The last step is linking the result in the tilde system with the initial one. The relationship can be easily obtained through equation 7.2.3 and the properties of variance.

$$\begin{aligned}
P(t) = Var[x(t)] &= Var[U\tilde{x}(t)] \\
&= UVar[\tilde{x}(t)]U^T \\
&= U\tilde{P}(t)U^T
\end{aligned} \tag{7.2.17}$$

It is possible to invert the previous relationship to get:

$$\tilde{P}(t) = U^{-1}P(t)U^{-T} \tag{7.2.18}$$

and, in particular:

$$\tilde{P}_0 = \tilde{P}(0) = U^{-1}P(0)U^{-T} = U^{-1}P_0U^{-T} \tag{7.2.19}$$

Once the $P(t)$ matrix is known, it is possible to apply the output transformation to get the variance of the output y:

$$Var[y(t)] = Var[Cx(t)] = CVar[x(t)]C^T = CP(t)C^T \tag{7.2.20}$$

The whole procedure may be summarized as follows.

Algoritmo 7.1 General algorithm to solve a time-varying differential Lyapunov matrix equation.

1. Take a linear stochastic system

$$\begin{cases} \frac{dx(t)}{dt} = Ax(t) + Bu(t) \\ y(t) = Cx(t) \end{cases} \tag{7.2.21}$$

where A is diagonalizable and $Var[x(0)] = P_0$.

2. Find an invertible U and a diagonal $\tilde{A}$ such that $\tilde{A} = U^{-1}AU$ and call $\tilde{a}_i$ the i-th element on its diagonal.

3. Define $\tilde{P}_0 = U^{-1}P_0U^{-T}$ and $\tilde{B} = U^{-1}B$.

4. Compute

$$\tilde{P}_{ij}(t) = \begin{cases} \left\{\tilde{P}_0\right\}_{ij} e^{(\tilde{a}_i+\tilde{a}_j)t} + \frac{1}{\tilde{a}_i+\tilde{a}_j}\left\{\tilde{B}S\tilde{B}^T\right\}_{ij}\left(e^{(\tilde{a}_i+\tilde{a}_j)t} - 1\right) & \tilde{a}_i \neq 0 \vee \tilde{a}_j \neq 0 \\ \left\{\tilde{P}_0\right\}_{ij} e^{(\tilde{a}_i+\tilde{a}_j)t} + \left\{\tilde{B}S\tilde{B}^T\right\}_{ij} t & \tilde{a}_i = \tilde{a}_j = 0 \end{cases} \tag{7.2.22}$$

5. Get back to $P(t) = U\tilde{P}(t)U^T$.

6. Apply the output transformation to get $Var[y(t)] = CP(t)C^T$, if required.

7.3 A simple test case: deriving the variance of an LMR-GW model

To put the algorithm to the test, a simple case is considered.

A single-underlying LMR-GW is transformed into a stochastic linear systems and the computations described in Algorithm 7.1 are run. The solution to this problem is known, as it was derived and proved in Chapter 4. The results given by the new algorithm are thus expected to coincide with the ones presented in Equation 4.3.14.

The matrices of the second-order stochastic system to consider were derived in Section 3.5:

$$A = \begin{bmatrix} -\lambda & \lambda \\ 0 & 0 \end{bmatrix} \tag{7.3.1}$$

$$B = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \tag{7.3.2}$$

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix} \tag{7.3.3}$$

Moreover, the covariance matrix of the white noise is

$$Cov[u(t_1), u(t_2)] = S\delta(t_1 - t_2) = I_2\delta(t_1 - t_2), \ \forall t_1, t_2 > 0 \tag{7.3.4}$$

and the initial variance P_0 is considered a generic symmetric matrix:

$$P_0 = \begin{bmatrix} p_1 & p_{12} \\ p_{12} & p_2 \end{bmatrix} \tag{7.3.5}$$

All the inputs required to run Algorithm 7.1 have been defined. The steps of the procedure are now followed.

The matrix U and $\tilde{A}$ were found in Chapter 4 to be:

$$U = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \tag{7.3.6}$$

$$\tilde{A} = \begin{bmatrix} -\lambda & 0 \\ 0 & 0 \end{bmatrix} \tag{7.3.7}$$

Now it is possible to compute the $\tilde{P}_0$ matrix:

$$\tilde{P}_0 = U^{-1} P_0 U^{-T} = \begin{bmatrix} p_1 - 2p_{12} + p_2 & p_{12} - p_2 \\ p_{12} - p_2 & p_2 \end{bmatrix} \tag{7.3.8}$$

and the auxiliary matrix $\tilde{B}$:

$$\tilde{B} = U^{-1} B = \begin{bmatrix} \sigma_1 & -\sigma_2 \\ 0 & \sigma_2 \end{bmatrix} \tag{7.3.9}$$

One has now all the ingredients required to compute the elements of matrix $\tilde{P}$:

$$\begin{aligned} \tilde{P}_{11} &= (p_1 - 2p_{12} + p_2)\, e^{-2\lambda t} - \frac{\sigma_1^2 + \sigma_2^2}{2\lambda} \left(e^{-2\lambda t} - 1 \right) \\ \tilde{P}_{12} = P_{21} &= (p_{12} - p_2)\, e^{-\lambda t} + \frac{\sigma_2^2}{\lambda} \left(e^{-\lambda t} - 1 \right) \\ \tilde{P}_{11} &= p_{22} + \sigma_2^2 t \end{aligned} \tag{7.3.10}$$

The last step is now to apply the transformation to get back to the P matrix, that is the time-varying variance of the original states:

$$P = U \tilde{P} U^T = \begin{bmatrix} P_{11}(t) & P_{12}(t) \\ P_{21}(t) & P_{22}(t) \end{bmatrix} \tag{7.3.11}$$

where:

$$\begin{aligned} P_{11}(t) &= \left(p_1 - 2p_{12} + p_2 - \frac{\sigma_1^2 + \sigma_2^2}{2\lambda} \right) e^{-2\lambda t} + 2 \left(p_{12} - p_2 + \frac{\sigma_2^2}{\lambda} \right) e^{-\lambda t} + \sigma_2^2 t + \frac{\sigma_1^2 - 3\sigma_2^2}{2\lambda} + p_2 \\ P_{12}(t) = P_{21}(t) &= \left(p_{12} - p_2 + \frac{\sigma_2^2}{\lambda} \right) e^{-\lambda t} + \sigma_2^2 t + p_2 - \frac{\sigma_2^2}{\lambda} \\ P_{22}(t) &= \sigma_2^2 t + p_2 \end{aligned} \tag{7.3.12}$$

This is exactly the same result derived in Chapter 4, Equation 4.3.14.

7.4 Value at Risk

The Value at Risk is defined as the maximum amount of money a portfolio can loose in a given time range in the p percent of the most favourable cases. Thus, given the cumulative distribution of the returns of the whole portfolio, the VaR is the $(1-p)$ percentile. To better appreciate the idea, the following figure provides a graphical interpretation.

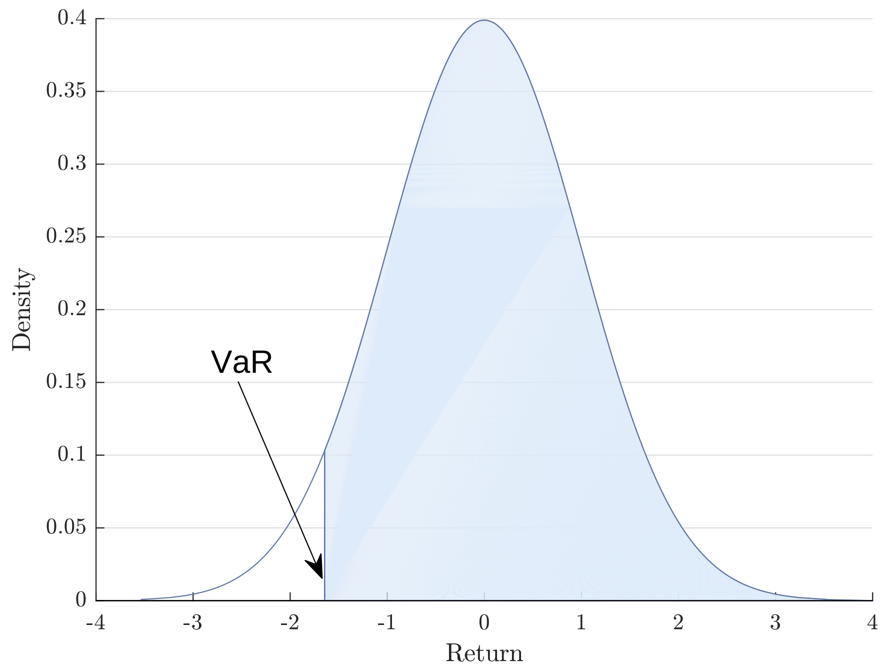


Figure 7.4.1: Value at Risk in the distribution of returns.

VaR was first introduced at the beginning of '90s as a consequence of the market cash of 1987. Its usefulness and reliability were sometimes severely criticized, mainly for two reasons:

- It does not account for the amount of the possible loss in the $(1-p)$ most unfavourable events
- It does not take into account very unlikely events.

In a famous speech in 2008 [11], David Einhorn compared the VaR to "an airbag that works all the time, except when you have a car accident".

Still, VaR is nowadays a de facto standard and its adoption is suggested by the Basel II and Basel III agreements, worldwide regulations on financial institutions [18].

The challenging part in computing VaR is inferring the probability distribution of the returns. The following section describe the most common approaches.

7.4.1 How to compute the VaR

Ideally, the correct probability distribution of the return should be extracted from the future behaviour of the asset composing the portfolio. This is not possible, as the future behaviour of the asset is unknown. So, the problem of computing VaR reduces to the problem of finding the best possible approximation for the future distribution of returns. Three approaches are currently employed:

7.4.1.1 Parametric approach

Prices are assumed to follow a Geometric Brownian Motion. The distribution of the returns is thus Gaussian. This assumption is very common in finance: it complies, for example, with the Black-

Scholes model. The mean is assumed to be zero and the covariance matrix P is estimated from the time series of the assets. As the variance of a GBM scales as t, at a specific time in the future, the multivariate distribution of the returns of each asset is $N(0, Pt)$.

Now, assume the overall return of the portfolio is a linear combination of the returns of its assets. Name w the column vector of the weights of each asset. It is possible to compute the variance of the overall portfolio P_p as:

$$P_p(t) = w^T (P \cdot t) w \tag{7.4.1}$$

The VaR is then the $(1-p)$ percentile of a random variable with a Gaussian distribution $N(0, P_p)$.

Despite being very simple and easily readable, the parametric approach has several downsides. The most important are:

- The assumption on the distribution of returns is not always verified in practice
- The assumption on the linear dependence of the portfolio on its assets excludes any non-linear payoff, thus making the approach incompatible with most of derivatives, including options.

Actually, the latter limitation may be overcome with an approximation of the non-linear dependence of the derivative on the asset. The simplest approximation is the following.

Let S be the asset price and X the derivative price. Then,

$$dX = \frac{\partial X}{\partial S} dS \tag{7.4.2}$$

The partial derivative of the derivative price on the underlying price is called delta. For vanilla options, closed formulas are available to compute delta. In other cases, approximations are required - see, for example, [17]. Consider then an n-asset portfolio and let δ_i be the delta of the i-th asset, if this is a non-linear derivative, 1 otherwise. Define $\tilde{w}$ as the new weight vector whose i-th element is $\tilde{w}_i = w_i \delta_i$. Then, one can easily show that the approximated covariance matrix is

$$\tilde{P}_p(t) = \tilde{w}^T (P \cdot t) \tilde{w} \tag{7.4.3}$$

The VaR computed from $\tilde{P}_p(t)$ is an approximation which does not take into account the full non-linear dependence between underlyings and derivatives. It thus differs from the result one could get if the real relationships were considered.

7.4.1.2 Historical estimation

Historical estimation assumes that past returns are representative of the future distribution. The value of the portfolio is evaluated for a large number of past time instants, usually each day for several years.

The overall value of the portfolio changes as a function of the assets, thus returns can be computed. From the returns, the distribution is inferred and its $(1-p)$ percentile is directly extracted.

Historical estimation overcomes the problems of the parametric approach, as no assumption is made on both the distribution of the returns and the contribution of each asset to the overall returns of the portfolio. However, this approach relies on the big assumption that the past behaviour of the assets is representative of their future evolution.

7.4.1.3 Monte Carlo simulation

As described in Section 5.5, Monte Carlo (MC) methods can be employed to simulate a huge number of possible paths for an asset. The same idea can be applied to a whole portfolio. At each step of each simulation, assets evolve following their models. The whole portfolio is valued, thus leading to a multiple paths for the value of the whole portfolio.

MC methods require models for the assets and models require calibration of their parameters. Through the calibration on historical time series, models encode the information about the past evolution of the asset, but the random nature of the simulation allows to take into account a wider range of scenarios than the mere historical estimation. As in the historical estimation, no limit is imposed on how the assets contribute to the value of the portfolio.

Despite being introduced last, MC methods are rapidly catching up the historical approach: a McKinsey report published in 2012 [9] estimated 15% of major financial institution use Monte Carlo simulation, but “the figure is rapidly increasing” as “the Monte Carlo method is widely considered the better theoretical approach to simulation of risk”.

7.5 VaR and Lyapunov equation

The procedure for the computation of the variance of an arbitrary linear stochastic system enables to derive a parametric VaR while using a more complex, two-factor model for the assets.

As the computation of VaR is not a pricing problem, models should be calibrated against history, as for the Monte Carlo approach.

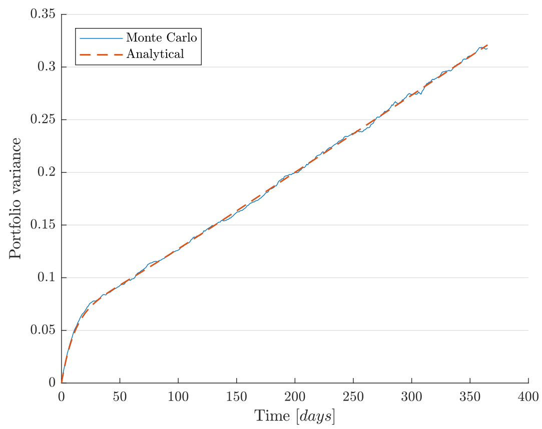
Chapter 2 showed how a model for the log-spot prices is by all means a model for the returns: the variance of its output is thus the variance of the returns itself. As the white noise driving the system is Gaussian, the state has, at each time instant, a Gaussian distribution. Assuming a zero mean, the variance is the only meaningful parameter for the distribution.

Algorithm 7.1 can thus lead to the variance P of the returns of each asset in the portfolio, at each time instant. When a two-factor model is used, weights should be carefully chosen to nullify the contribution of the terms related to long-term trends. The long-term trends are indeed internal states of the model and do not represent the value of any real asset, as the price of the assets is given by the short-term components. Then, the covariance matrix of the returns on the overall portfolio P_p is computed as in the parametric approach. Let w be the column vector of the weights of each asset.

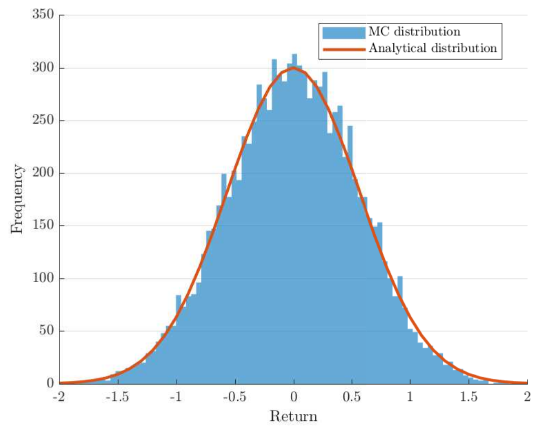
$$P_p\left(t\right) = w^T P\left(t\right) w \tag{7.5.1}$$

The final step, as in the parametric approach, is the computation of the $(1-p)$ percentile of the random variable $N\left(0, P_p\right)$.

Even if it resembles the parametric approach, the result is exactly equivalent to a Monte Carlo approach which leverages the same model. The equivalence between MC simulation and the analytical result was already shown in Section 6.4 for a model with two underlyings. The following figure compares the variances of a portfolio with 4 assets. Each of the assets is modelled by a two-factor LMR-GW model and correlations are not negligible.



(a) Variance over time: MC vs analytical approach.



(b) Distribution of returns: MC vs analytical.

Figure 7.5.1: Monte Carlo approach vs analytical approach.

It is evident that the two results coincide except for the error introduced by the random nature of MC simulation. The time evolution of the variance is almost overlapped and the Gaussian distribution derived from the analytical variance almost perfectly fits the histogram of MC outcomes.

The analytical approach to the VaR is an intermediate solution between the parametric solution and Monte Carlo simulation. It is as simple and fast to compute as the parametric approach, but the variance is determined by the model itself, like in the MC method. On the other hand, it inherits the limitations of the parametric approach on the assets. Only the VaR of a portfolio made of asset which can be modelled by a linear stochastic systems can be exactly computed. When non-linear derivatives are involved, approximations are required.

Part IV

Conclusion

Chapter 8

Final remarks and future developments

The main goal of this work was pricing a European vanilla option on an energy commodity.

The problem of pricing a vanilla option was brilliantly solved in 1973, when Black and Scholes published the study which earned them the Nobel prize. As powerful as it is, Black-Scholes model relies on the knowledge of the volatility of the underlying, which is hard to estimate. As pricing is about being in line with the market, the best way to infer volatility is to fit the model to market prices.

Classical models are easy to calibrate, but they may not be effective in capturing the peculiarities of the assets listed in energy markets. Two-factor models are more effective, but their calibration may be computationally expensive. By transforming a two-factor model into a linear stochastic system, a closed-form solution for the volatility of an asset is derived, thus drastically reducing the computational effort required by calibration.

Two-factor models were calibrated against both market prices and historical time series. Models were then asked to price options other than those used in the calibration. Two-factor models calibrated against the market outperformed both single-factor models with the same calibrations and two-factor models calibrated against the history.

The encouraging results on vanilla options led to the application of the same ideas to models including more than one asset. A procedure to price spread options was developed and tested against the state-of-the-art Monte Carlo approach. The procedure showed the same results as MC simulation, with a remarkable decrease in computation time.

In the end, a general algorithm to compute the variance of a linear stochastic system is derived. The algorithm provides an analytical expression for each term of the covariance matrix under the hypothesis that the dynamic matrix of the system is diagonalizable. The result is applied to compute the Value at Risk of a generic portfolio, where all the assets can be modelled by a linear stochastic system. The result is compared with Monte Carlo simulation: outcomes are almost identical.

The application of control system theory to finance is not new in the literature. Kalman filter was applied to accomplish many tasks, including the calibration of stochastic models for financial assets against price history. However, the idea of leveraging the Lyapunov equation to derive the variance of an asset was rarely followed.

This work is very far from being an exhaustive study. Models other than the LMR-GW may be tested and other pieces of the standard financial theory may be included. Most notably, models with stochastic volatility may be fitted into the stochastic linear systems framework. Moreover, more tests may be carried out to investigate the origin of the difference in performance between single-factor and two-factor models.

Improvements are also possible in the field of market calibration. For instance, an in-depth study of the most appropriate loss function may be carried out.

Acknowledgements

First things first, I have to thank you, my reader.

I hope something in the pages you read has caught your attention, made you look for more details or, simply, made you think. I tried to be as rigorous as I must, as clear as I can, as interesting as everyone should be. If I failed in any of these goals, it is my fault, only my fault, completely my fault. My apologies.

My gratitude and respect to the ones who made this work possible. Some month ago I barely knew what a derivative is. Learning is as beautiful as difficult, good teachers make it way easier. It doesn't matter if teaching is their job or not. Maybe you know, maybe you don't: you gave me lots, lots more than what is written on these pages.

I cannot forget that two of the jobs I got - on two - came from my teachers. Thank you for believing in me, even when you did not have to, even if you had no reason to.

Not only teachers, here. My ever-lasting gratitude goes to my students: if you learnt from me a tenth of what I learnt from you, I am a better teacher than I believe to be.

Now, personal matters are coming.

Little of what happen to me in the last three years would have been possible without a five-minute chat. I have to thank the guy I talked with. You changed everything - I mean, every single thing - and for the better, mostly.

A beach, a football field, the office of an architect. What is the common element? Wonderful people made them host some of the best moments in my life. Few of them will ever read this page, but I have to write these words anyway. I would not be honest, otherwise.

And even when I was far from that places, I was so lucky to find someone who made me feel at home. When home was a thousand miles far.

I've mentioned two jobs. I expected to find colleagues, I found friends. I was introduced bosses, I knew mentors. Thank you all, and thanks to the friends who are going to become my colleagues.

In the end, a great thank to three little girls. I don't have to state why. And this is a reason itself.

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